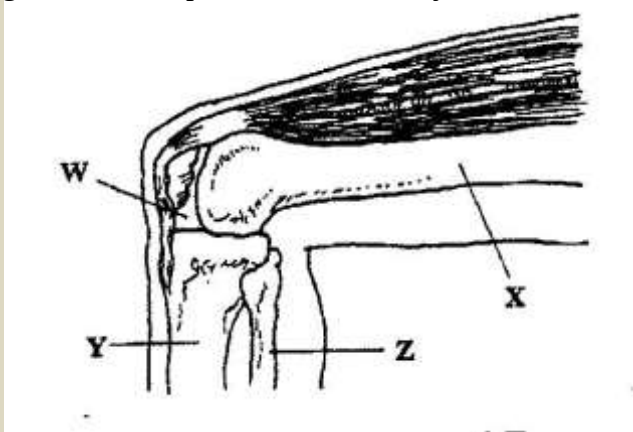


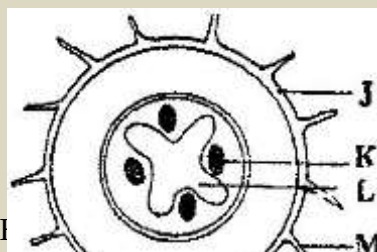
K.C.S.E PAPER 231/2 2006

1. The diagram below represents bones at a joint found in the hind limb of a mammal.



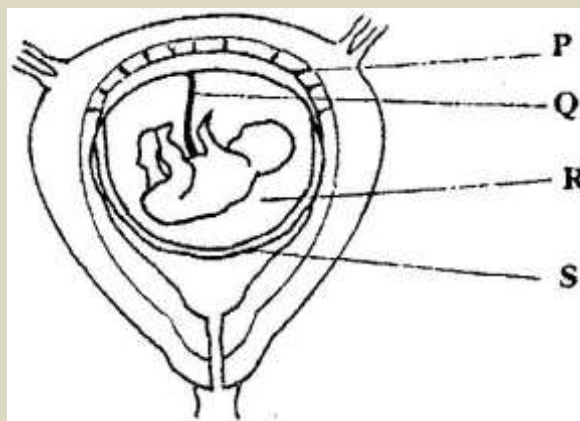
- a) Name the bones labeled XY and Z (3mks)
- b) i) Name the substance found in the place labeled W. (1mk)
ii) State the function of the substance named in (b) (i) above. (1mk)
- c) Name the structure that joins the bones together at the joint. (1mk)
- d) State the differences between ball and socket joint and the one illustrated in the diagram above. (1mk)
- e) Name the structure at the elbow that performs the same function as the same function as the patella. (1mk)
2. a) Name two disorder in human caused by gene mutation. (2mks)
- b) Describe the following chromosomal mutations. (2mks)
a. Inversion
b. Translocation.
- c) In mice the allele for black fur is dominant to the allele for brown fur.
What percentage offspring would have brown fur form across between heterozygous black mice? Show your working.
Use letter B to represent the allele for black colour. (4mks)
3. a) Distinguish between pyramid of numbers and pyramid of biomass. (2mks)
- b) Give three reasons for loss of energy from one trophic level to another in the food chain. (3mks)

4. The diagram below represents a traverse section through a plant organ



- A. From which plant organ was the section obtained? (1 mk)
- B. Give two reasons for your answer in (a) above. (2mks)
- C. Name the parts labeled J,K and L. (3mks)
- D. State two functions of the part labeled M. (2mks)

5. The diagram below represents human foetus in a uterus.



- a) Name the part labeled S. (1mk)
- b) i) Name the types of blood vessels found in the structure labeled Q. (2mks)
- ii) State the differences in composition of blood found in the vessels named in (b)(i) above. (2mks)
- c) Name two features that enable the structure labeled P carry out its function. (2mks)
- d) State the role of the part labeled R (1mk)

SECTION B

Answer question 6 (compulsory) in the spaces provided and either question 7 or 8 in the spaces provided

6. An experiment was carried out to investigate the effect of hormones on growth

KCSE BIOLOGY PAPER 2 2006-2019 (MR CHEPKWONY)

of lateral buds of three pea plants

The shoots were treated as follows:

Shoot A – Apical bud was removed.

Shoot B – Apical bud was removed and gibberellic acid placed on the cut shoot.

Shoot C – Apical bud was left intact.

The length of the branches developing from lateral buds were determined at Regular intervals.

The results obtained are as shown in the table below.

Time in days	Length of branches in millimeters		
	Shoot A	Shoot B	Shoot C
0	3	3	3
2	10	12	3
4	28	48	8
6	50	9	14
8	80	120	20
10	118	152	26

- a) Using the same axes, draw graphs to show the lengths of branches against time. (8mks)
- b) i) What was the length of the branch in shoot B on the 7th day? (1mk)
- ii) What would be the expected length of the branch developing from shoot A on the 11th day? (1mk)
- c) Account for the results Obtained in the experiment (6mks)
- d) Why was shoot C included in the Experiment? (1mk)
- e) What is the importance of gibberellic acid in agriculture? (1mk)
- f) State two physiological processes that are brought about by the application of gibberellic acid on plants. (2mks)
7. Describe how human kidney functions (20mks)
8. Describe how water moves from the soil to the leaves in a tree. (20mks)

BIOLOGY

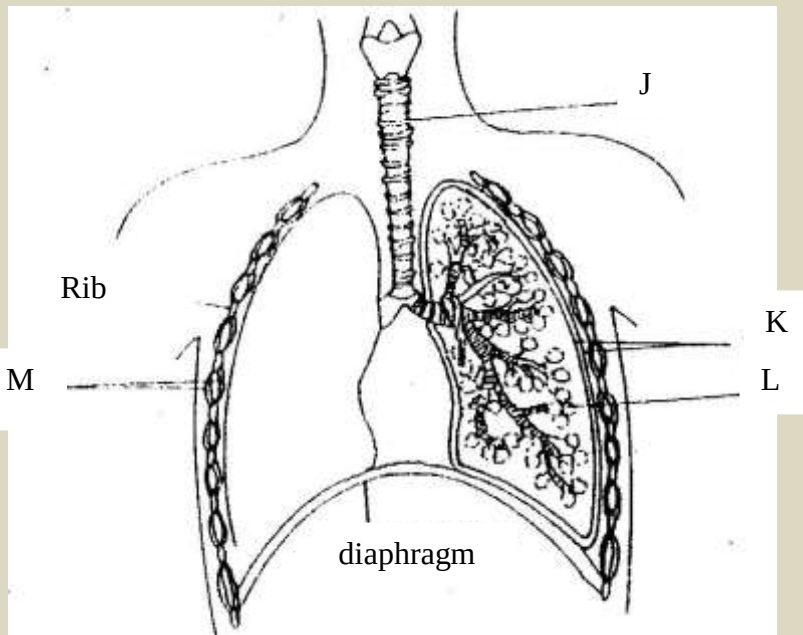
K.C.S.E PAPER 231/2 2007

QUESTIONS

SECTION A (40 mks)

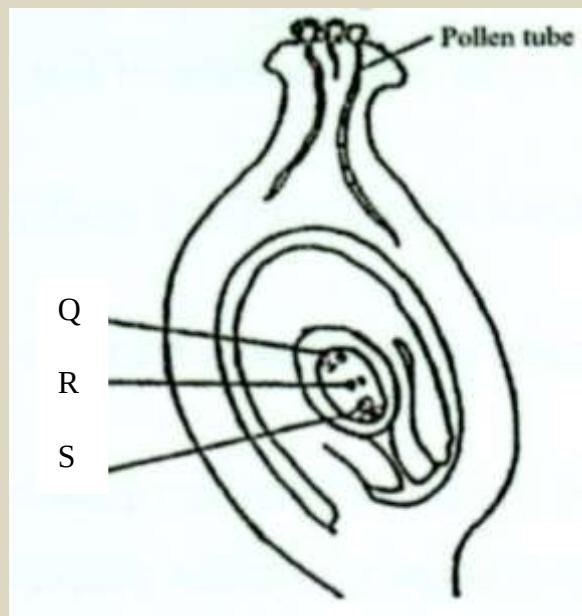
Answer all questions in this section in the spaces provided

1. The diagram below represents some gaseous exchange structures in humans



- (a) Name the structures labeled K, L, and M (3 mks)
K.....
L.....
M.....
- (b) How is the structure labeled J suited to its function? (3 mks)
- (c) Name the process by which inhaled air moves from the structure labeled L into blood capillaries (1 mk)
- (d) Give the scientific name of the organism that causes tuberculosis in humans (1 mk)
2. (a) Explain what happens to excess amino-acids in the liver of humans (3 mks)
- (b) Which portion of the human nephron are only found in the cortex? (3 mks)
- (c) (i) What would happen if a person produced less antidiuretic hormone? (1 mk)
(ii) What term is given to the condition described in (c) (i) above (1 mk)
3. (a) What is meant by the following terms (1 mk)
(i) Protandry (1 mk)
(ii) Self sterility? (1 mk)
- (b) The diagram below shows a stage during fertilization in a plant

Pollen Tube



- (i) Name the parts labeled Q, R, and S (3 mks)
- Q
R
S
- (ii) State two functions of the pollen tube (2 mks)
- (c) On the diagram label the micropyle (1 mk)
- 4 (a) Name the three type of muscles found in mammals and give an example of where each of them is found (3 mks)
- | Type of muscle | Where found |
|----------------|-------------|
| (i) | |
| (ii) | |
| (iii) | |
- (b) State the difference between ball and socket and hinge joint (1 mk)
- (c) State the functions of synovial fluid (2 mks)
- (d) State two advantages of having an exoskeleton (2 mks)
5. In maize the gene for purple colour is dominant to the gene for white colour. A pure breeding maize plant with purple grains was crossed with a heterozygous plant.
- (a) (i) Using letter G to represent the gene for purple colour, work out the genotype ratio of the offspring (5 mks)
- (ii) State the phenotype of the offspring (1 mk)
- (b) What is genetic engineering? (1 mk)
- (c) What is meant by hybrid vigour? (1 mk)

SECTION B (40 MKS)

KCSE BIOLOGY PAPER 2 2006-2019 (MR CHEPKWONY)

Answer questions 6 (compulsory) in the spaces provided and either questions 7 or 8 in the spaces provided after questions 8

6. In the experiment to determine the effect of ringing on the concentration of sugar in phloem a ring of bark from the stem of a tree was cut and removed. The amount of sugar in grammes per 16cm^3 piece of bark above the ring was measured over a 24 hour period. Sugar was also measured in the bark of a similar stem of a tree which was not ringed. The results are shown in the table below.

Time of the day	Amount of sugar in grammes per 16cm^3 piece of bark	
	Normal stem	Ringed stem
06 45	0.78	0.78
09.45	0.80	0.91
12.45	0.81	0.01
15 45	0.80	1.04
18.45	0.77	1.00
21 45	0.73	0.95
00 45	0.65	0.88

- (a) Using the same axes, plot a graph of the amount of sugar against time (6 mks)
- (b) At what time was the amount of sugar highest in the
- (i) Ringed stem (1 mk)
- (ii) Normal stem? (1 mk)
- (c) How much sugar would be in the ringed stem if it was measured at 0345 hours? (1 mk)
- (d) Give reasons why there was sugar in the stems of both trees at 06 45 hours (2 mks)
- (e) Account for the shape of the graph for the tree with ringed stem between:
- (i) 06 45 hours and 15 45 hours (3 mks)
- (ii) 15 45 hours and 00 45 hours (2 mks)
- (f) Name the structures in phloem that are involved in the translocation of sugars (2 mks)
- (g) Other than sugars name two compounds that are translocated in phloem (2 mks)
7. Describe the structure and functions of the various parts of the human ear (20 mks)
8. Describe causes and methods of controlling water pollution (20 mks)

BIOLOGY

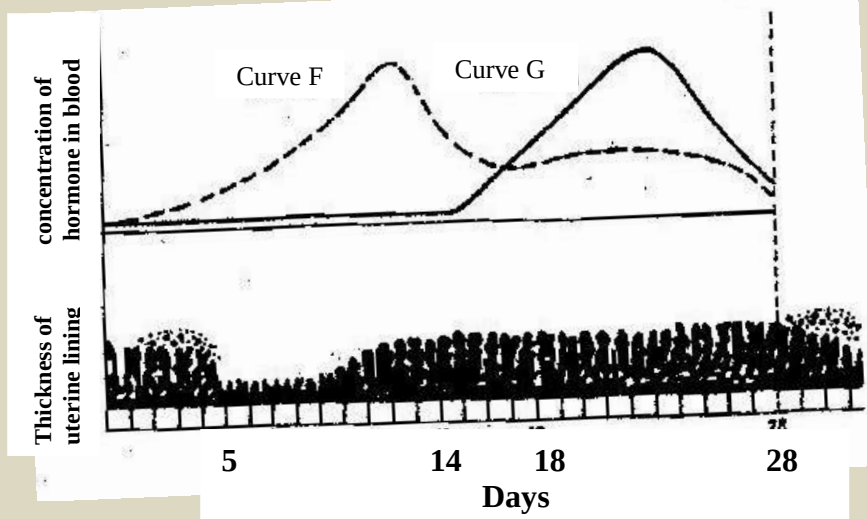
K.C.S.E PAPER 231/2 2008

QUESTIONS

SECTION A (40 MKS)

Answer all the questions in this section in the spaces provided

1. The figure shows changes that take place during menstrual cycle in human



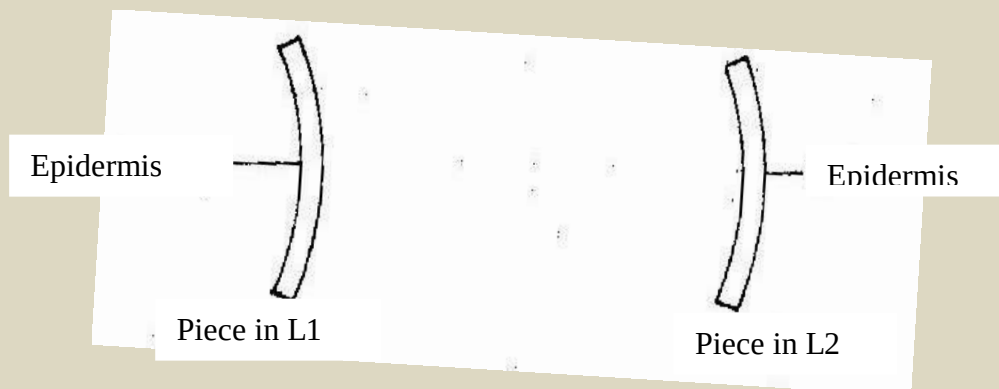
- a) Name the hormone whose concentrations are represented by curves F and G (2 mks)
- b) State the effects of the hormones named in (a) above on the lining of the uterus (2 mks)
- c) (i) Name the hormone which is released by the pituitary gland in high concentration on the 14th day of the menstrual cycle (1 mk)
- (ii) State two functions of the hormone named in (c) (I) above (2 mks)
- (d) State the fertile period during the menstrual cycle (1 mk)
2. A pea plant with round seeds was crossed with a pea plant that had Wrinkled Seeds the gene for round seeds is dominant over that for wrinkled seeds
- Using letter R to represent the dominant gene state:
- i) The genotype of parents if plant with round seed was heterozygous (2 mks)
- ii) The gametes produced by the round and wrinkled seed parents
- Round seed parent
- Wrinkled seed parent
- The genotype and phenotype of F₁ generation. Show your working (3 mks)

3.

KCSE BIOLOGY PAPER 2 2006-2019 (MR CHEPKWONY)

- a) What is a test – cross? (1 mk)
- b) The equation below represents a process that takes place in plants
$$6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$$
- (a) Name the process (1 mk)
- (b) State two conditions necessary for the process to take place (2 mks)
- (c) State what happens to the end- products of the process (5 mks)
4. (a) Give three reasons in each case why support is necessary in
- (i) Plants (3 mks)
- (ii) animals (3 mks)
- (b) Why is movement necessary in animals (2 mks)
5. A freshly obtained dandelion stem measuring 5 cm long was split lengthwise to obtain two similar pieces. The pieces were placed in solutions of different concentrations in Petri dishes for 20 minutes.

The appearance after 20 minutes is as shown



- a) Account for the appearance of the pieces in solutions L₁ and L₂ (6 mks)
- b) State the significance of the biological process involved in the experiment (2 mks)

SECTION B (40 Mks)

Answer question 6 (compulsory) and either questions 7 or 8 in the spaces provided

6. An experiment was carried out to investigate transpiration and absorption of water in sunflower plants in their natural environment with adequate supply of water. The amount of water was determined in two hour intervals. The results are as shown in the table below

Time of day	Amounts of water in grammes	
	Transpiration	Absorption
11 00 - 13 00	33	20
13 00 - 15 00	45	30
15 00 - 17 00	52	42
17 00 - 19 00	46	46
19 00 - 21 00	25	32
21 00 - 23 00	16	20
23 00 - 01 00	08	15
01 00 - 03 00	04	11

- a) Using the same axes, plot graphs to show transpiration and absorption of water in grammes against time of the day (7 mks)
- b) At what time of the day was the amount of water the same for transpiration and absorption? (1 mk)
- c) Account for the shape of graph of:
Transpiration (3 mks)
Absorption (3 mks)
- d) What would happen to transpiration and absorption of water if the experiment was continued till 05 00 hours? (2 mks)
- e) Name two factors that may affect transpiration and absorption at any given time (2 mks)
- f) Explain how the factors you named in (e) above affect transpiration (2 mks)
- g) Describe the nitrogen cycle (20 mks)
- (a) State four characteristics of gaseous exchange surfaces (4 mks)
- (b) Describe the mechanism of gaseous exchange in a mammal (16 mks)

BIOLOGY

K.C.S.E PAPER 231/2 2009

QUESTIONS

SECTION A (40 MKS)

Answer all the questions in this section in the spaces provided

1. When the offspring of purple and white flowered pea plants were crossed, they produced purple and white flowered plants in the ratio of 3: 1

Using letter H to represent the gene for purple colour

- (a) State the genotype of:

(i) Parents

(2 mks)

(ii) F₁ Generation

(1 mk)

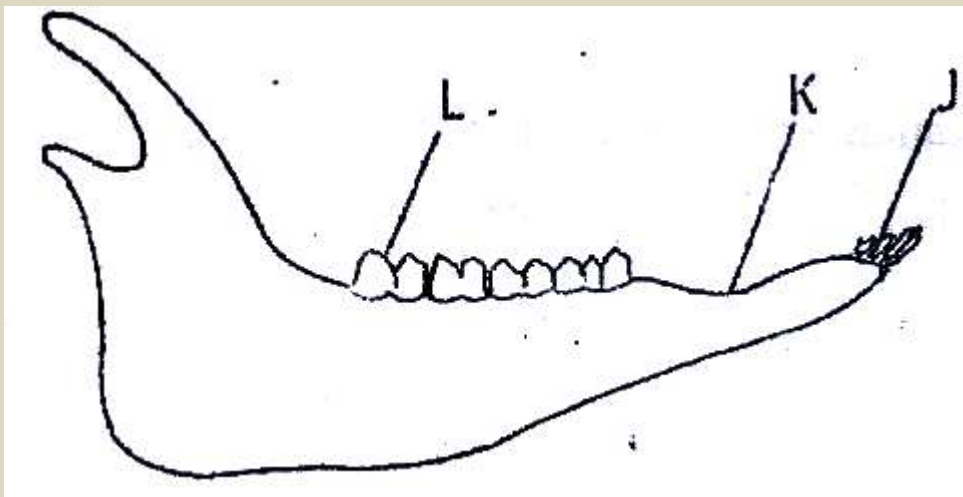
- (b) Work out the cross between plants in the F₁ generation

(4 mks)

- (c) Account for the colour the flowers in plants of the F₁ generation

(1 mk)

2. The diagram below represents the lower jaw of a mammal



- (a) Name the mode of nutrition of the mammal whose jaw is shown

(1 mk)

- (b) State one structural and one functional difference between the teeth labeled J and L

Structural

(1 mk)

Functional

(1 mk)

- (c) (i) name the toothless gap labeled K.

(1 mk)

- (d) Name the substance that is responsible for hardening of teeth

(1 mk)

KCSE BIOLOGY PAPER 2 2006-2019 (MR CHEPKWONY)

- 3.
- (a) i) what is meant by the term biological control (1 mk)
 - (ii) Give an example of biological control (1 mk)
 - (b) (i) What is eutrophication? (3 mks)
 - (ii) What are the effects of eutrophication (3 mks)
 - (c) Name a substance that is responsible for acid rain (1 mk)
- 4.
- (a)
 - (i) Explain the changes that take place in the pupil and iris of a human eye when a person moves from a dark room to a room with bright light (3 mks)
 - (ii) What is the significance of the changes explained in (a) above (1 mk)
 - (b) How does the human eye obtain nutrients? (3 mks)
 - (c) Explain why images that form on the blind spot are not perceived (2 mks)
- 5.
- (a) what happens when a wilting young plants is well watered (3 mks)
 - (b) Name a support tissue in plants thickened with
 - (i) Cellulose (1 mk)
 - (ii) Lignin (1 mk)
 - (c) Give three functions of pectoral and pelvic fins in a fish (3 mks)

SECTION B (40 MKS)

Answer questions 6 (compulsory) and either question 7 or 8 in the spaces provided

6. An experiment was carried out to investigate the effect of temperature on the rate of reaction catalyzed by an enzyme. The results are shown in the table below

Temperature ($^{\circ}\text{C}$)	Rate of reaction in mg of products per unit time
5	0.2
10	0.5
15	0.8
20	1.1
25	1.5
30	2.1
35	3.0
40	3.7
45	3.4
50	2.8
55	2.1

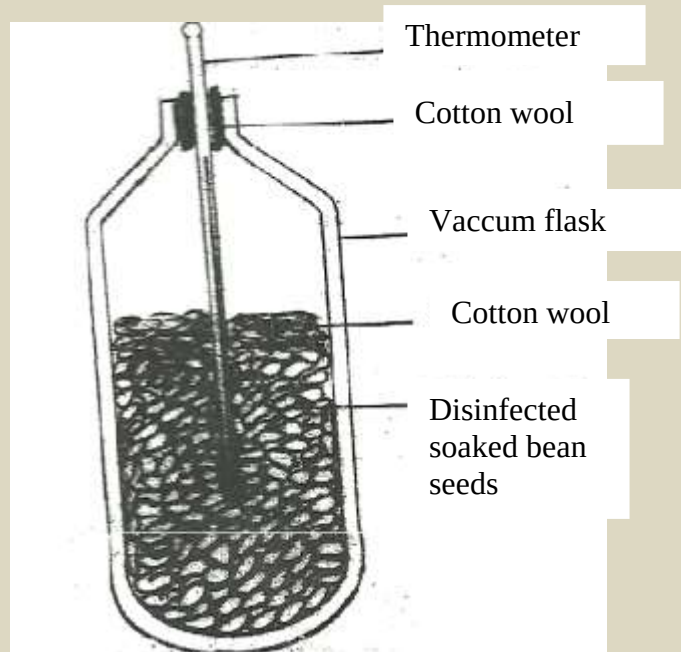
- a) On the grid provided draw a graph of rate of reaction against temperature (6 mks)
- (b) When was the rate of reaction 2.6 mg of product per unit time? (2 mks)
- (c) Account for the shape of the graph between
- (i) 5⁰ C and 40⁰ C (2 mks)
- (ii) 45⁰ C and 60⁰C (3 mks)
- (d) Other than temperature name two ways in which the rate of reaction between 5⁰C and 40⁰C could be increased (2 mks)
- (e) (i) Name one digestive enzymes in the human body which works best in acidic condition (1 mk)
- (ii) How is the acidic condition for the enzyme named in (e) (i) above attained? (2 mks)
- (f) The acidic conditions in (e) (ii) above is later neutralized
- (i) Where does the neutralization take place?
- (ii) Name the substance responsible for neutralization (1 mk)
7. How are flowers adapted to wind and insect pollination? (20 mks)
8. Describe the role of the liver in homeostasis in the human body (20 mks)

BIOLOGY
K.C.S.E PAPER 231/2 2010
QUESTIONS

SECTION A (40 MKS)

Answer all the questions in this section in the spaces provided.

1. In an experiment, disinfection soaked bean seeds were put in a vacuum flask which was then fitted with a thermometer as shown in the diagram below.

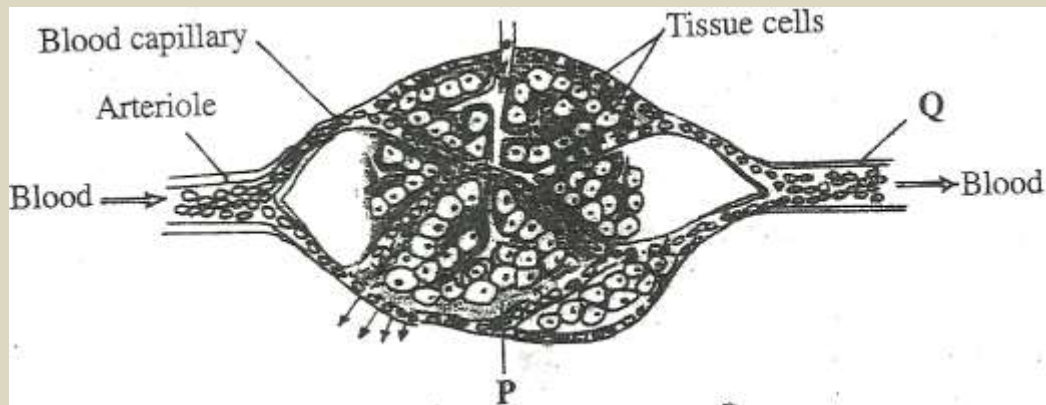


The temperature readings were taken every morning for three consecutive days.
Which process was being investigated? (1 mk)

- i) What were the expected results? (1 mk)
- ii) Account for the answer in (b) (i) above? (2 mks)
- Why were the seeds disinfected? (2 mks)
- Why was a vacuum flask used in the set-up? (1 mk)
- How would a control for this experiment be set? (1mk)

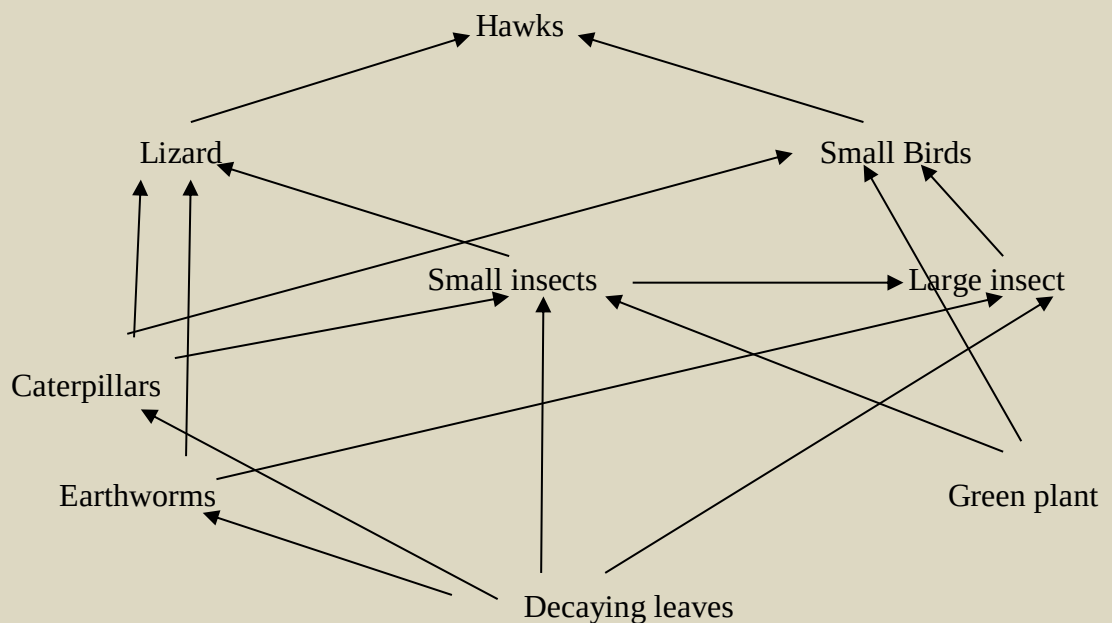
KCSE BIOLOGY PAPER 2 2006-2019 (MR CHEPKWONY)

2. The diagram below shows blood circulation in a mammalian tissue.



- a) Name the part labeled P and Q (2 mks)
- b) Name the substance that are:
 - c) Required for respiration that move out of capillaries; (2 mks)
 - d) Remove from tissue cells as a result of respiration (2 mks)
- e) Explain how substances move from blood capillaries into the tissue cells. (2 mks)
- f) Name one component of the blood that is not found in the part labeled P (1 mk)

3. The diagram below represents a food web in certain ecosystem



- a) Name the trophic level occupied by each of the following:
 - Caterpillars (1 mk)
 - Small insect. (1 mk)

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b) From the food web, construct two food chains which end with lizards as a tertiary consumer.

(2 mks)

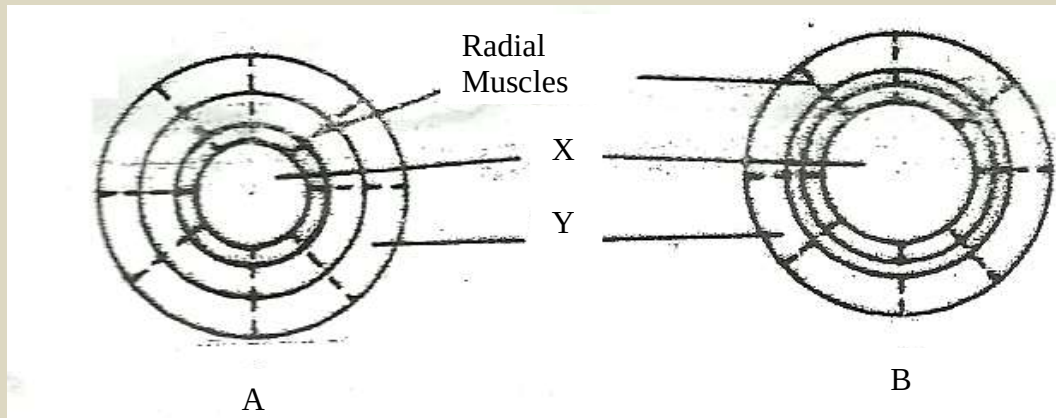
i) Which organisms have the least biomass in this ecosystem?

(1 mk)

ii) Explain the answer in (c)(i) above.

(3 mks)

4. The diagram below shows how the iris and pupil of a human eye appear under different conditions.



Name the structures labeled X and Y

(2 mks)

i) State the condition that leads to the change in appearance shown in the diagram labeled B

(2 mks)

ii) Describe the change that lead to the appearance of the iris and pupil as shown in the diagram labeled B.

(4 mks)

iii) What is the significance of the change described in (b) (ii) above?

(1 mk)

When pure breeding black guinea pigs were crossed with pure breeding white guinea pigs, the offspring had a coat with black and white patches.

Using letter G to represent the gene for black coat colour and letter H for white coat colour, work out the genotypic ratio of F₂.

State the phenotypic ratio of F₂.

(1 mk)

i) Name the term used when two alleles in heterozygous state are fully expressed phenotypically in an organism.

(1 mk)

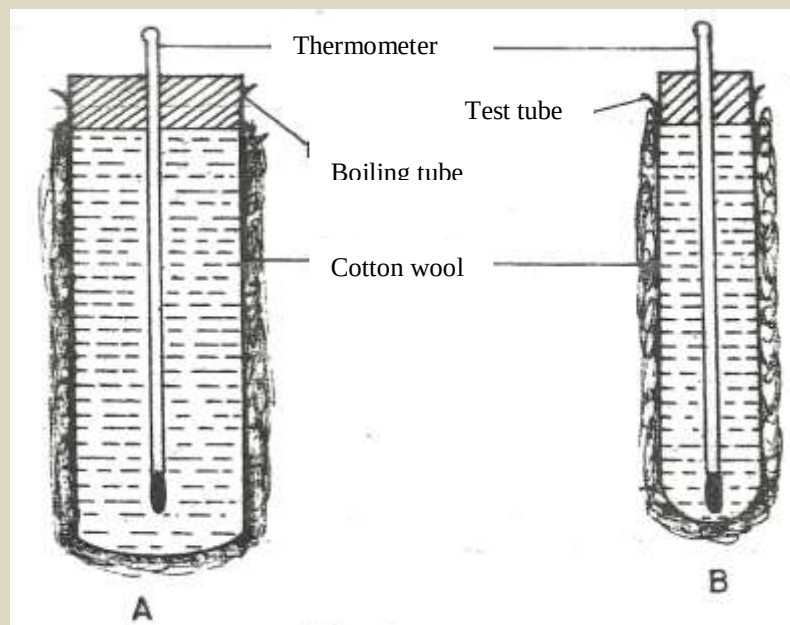
ii) Give an example of a trait in human beings where the condition whose term is named in (c) (i) above expresses itself.

(1 mk)

SECTION B

Answer question 6 (compulsory) and either question 7 and 8 in the space provided

6. In an experiment to investigate a certain physiological process, a boiling tube labeled A and a test tube labeled B were covered with cotton wool. The two tubes were simultaneously filled with hot water and fitted with thermometers. The experimental set-up was as in the diagrams below.



The temperatures reading were taken at the start and after every two minutes for twenty minutes. The results were as shown in the table below.

Time (minutes)	Temperature ($^{\circ}\text{C}$)	
	Boiling tube A	Test tube B
0	60	60
2	59	54
4	57	50
6	55	46
8	53	43
10	52	40
12	51	37
14	49	35
16	48	33
18	47	32
20	46	30

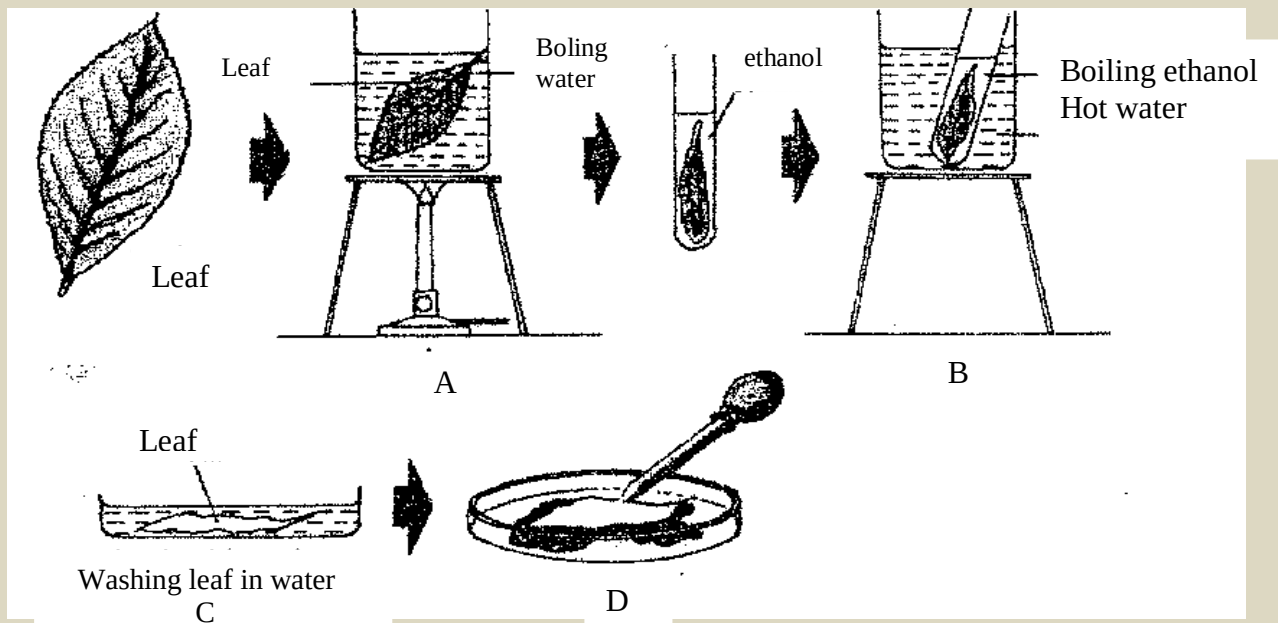
Using the same axes, draw graphs of temperature against time.

(6 mks)

- i) Work out the rate of heat in the boiling tube labeled A and test tube labeled B between the 5th and 15th minutes.
- ii) Account for the answers in (b) (i) above. (2 mks)
- iii) How does the explanation in (b) (ii) above apply to an elephant and a rat? (2 mks)
- i) State the role of the cotton wool in this experiment. (1 mk)
- ii) Name two structures in mammals that play the role stated in (c) (i) above (2 mks)
- State three advantage of having constant body temperature in mammals. (3 mks)
7. Describe the process of fertilization in flowering plants. (20 mks)
8. Describe how a finned fish such as Tilapia moves in water. (20 mks)

BIOLOGY
K.C.S.E PAPER 231/2 2011
QUESTIONS

1. The set-up below illustrates a procedure that was carried out in the laboratory with a leaf plucked from a green plant that had been growing in sunlight.



2.

What was the purpose of the above procedure ?

(1mk)

Give reasons for carrying out step A,B and C in this procedure.

(3mks)

(iii) Name the reagent that was used at the step labeled D .

(1mk)

(iv) State the expected result on the leaf after adding the reagent named in (iii) above. Stain dark blue/ Blue dark

In humans, hairy ears is controlled by agene on the Y Chromosomes .

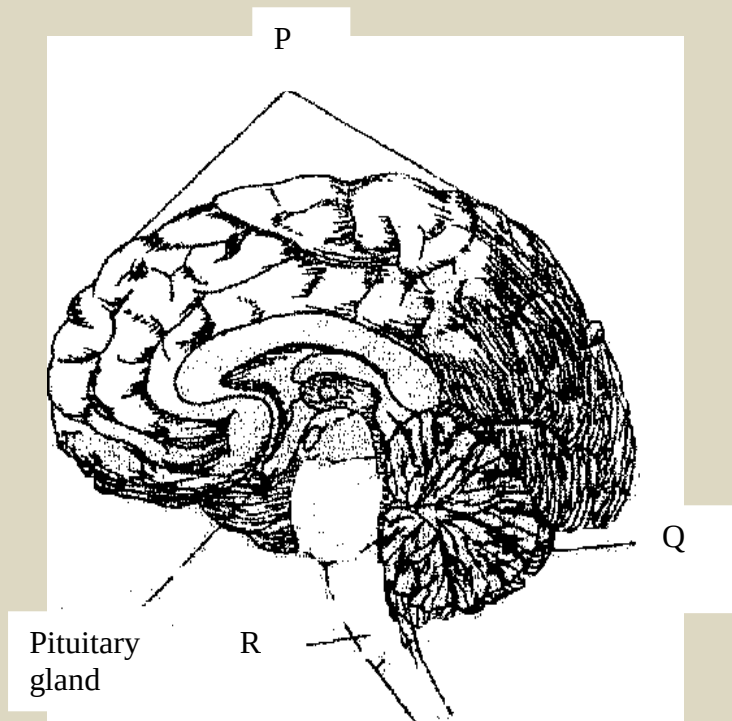
Using letter Y^H to represents the chromosome carrying the gene for hairy ears, work out a cross between a hairy eared man and his wife.

(4 mks)

(b) (i) What is the probability of the girls having hairy ears?

(1 mk)

- (ii) Give reason for your answer in (b) (i) above. (1mk)
- (c) Name two disorders in humans that are determined by sex linked genes (2mks)
- (d) Explain how comparative embryology is an evidence for organic evolution. (2mks)
3. (a) Name the causative agent for the following respiratory diseases. (2mks)
- (i) Whooping cough.
- (ii) Pneumonia
- (b) Describe how oxygen in the alveolus reaches the red blood cells. (4mks)
- (c) How are the pneumatophores adapted to their function? (2mks)
4. (a) the diagram below represents a section of the human brain.



- Name the structure labeled P and R.
State two functions of the part labeled Q
- (b) (i) Name two reproductive hormones secreted by the pituitary gland in women. (2mks)
- (ii) State one function of each of the hormones named in (b)(i) above (2mks)

5. (a) The diagram below represents a flower.

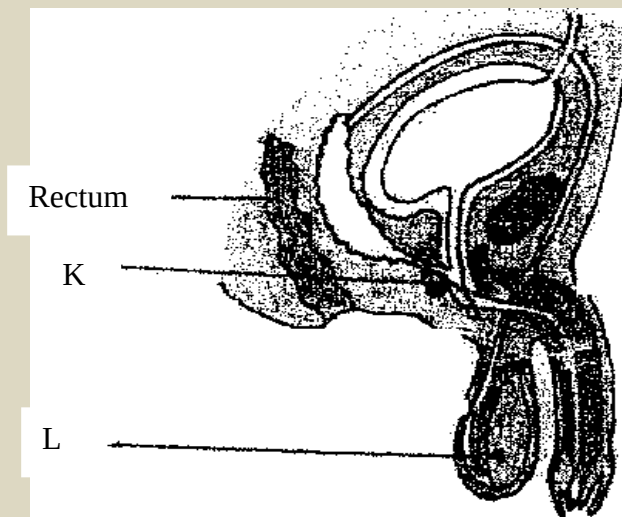


(i) On the diagram, name two structures where meiosis occurs.

(2 mks)

(ii) How is the flower adapted to prevent self-pollination?

(b) The diagram below represents a human reproductive organ.



(i) Explain two adaptations of the structure labeled L to its functions

(2 mks)

Explain the role of gland labeled K

SECTION B (40 MKS)**Answer 6(compulsory) and either question 7 or 8 in the spaces provided**

6. (a) An experiment was carried out to investigate the population of a certain micro-organism.

Two petri-dishes were used. Into the petri-dish labeled M, 60cm³ of a culture medium was placed while 30cm³ of the same culture medium was placed in petri-dish labeled N. Equal numbers of the micro-organisms were introduced in both petri-dishes. The set-ups were then incubated at 35°C. The number of micro-organisms in each petri-dish was determined at irregular intervals for a period of 60 hours. The results were as shown in the table below

Relative number of micro-organisms	M	40	40	180	280	1200	1720	1600	1840	1560	600
	N	40	40	120	200	680	560	560	600	600	400
Time in hours		0	5	10	15	23	30	35	42	45	60

- i) On the same axes, draw the graphs of relative number of micro-organisms against time on the grid provided. (7 mks)
- (ii) After how many hours was the difference between the two populations greatest? (1 mk)
- (iii) Work out the difference between the two populations at 50 hours (2 mks)
- (vi) With a reason state the effect on the population of micro-organisms in petri-dish M if the temperature was raised to 60°C after 20 hours. (2mks)
- (v) Account for the shape of the curve for population in petri-dish N between 46 hours and 59 hours. (3mks)
- (b) Explain how osmotic pressure in the human blood is maintained at normal level. (5mks)

7. (a) Explain how structural features in terrestrial plants affect their rate of transpiration. (13 mks)

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- (b) Explain how the human skin brings about cooling of the body on a hot day . (7mks)
8. (a) Describe the exoskeleton and its functions in insects . (13mks)
- (b) Describe how accommodation in the human eye is brought about when focusing on a near object . (7mks)

BIOLOGY

K.C.S.E PAPER 231/2 2012
QUESTIONS

SECTION A (40 mks)

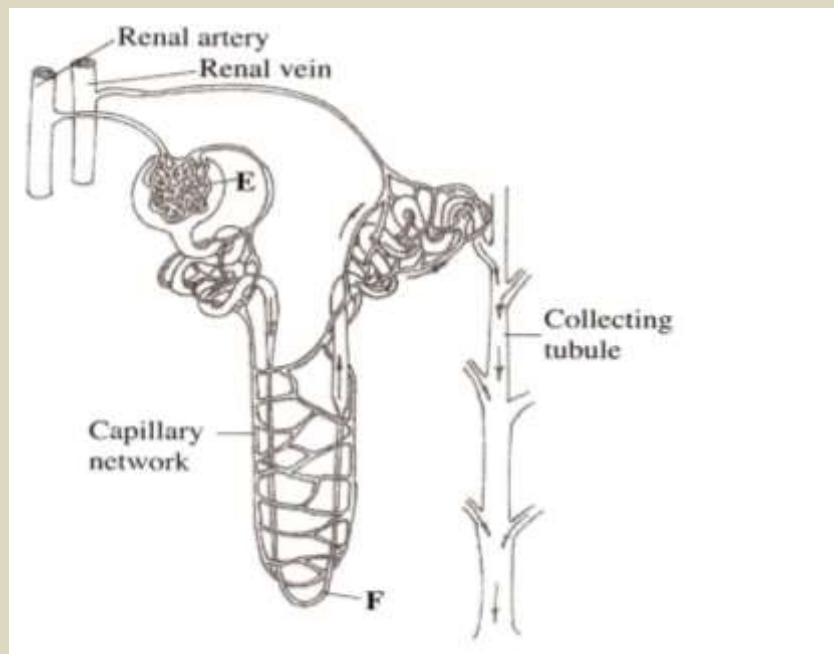
Answer ALL the questions in this section in the spaces provided.

1. In a certain plant species which is normally green, a recessive gene for colour (n) causes the plants to be white in colour. Such plants die at an early age. In the heterozygous state, the plants are pale green in colour but grow to maturity.

Give a reason for the early death of the plants with the homozygous recessive gene. (2 mks)

- (b) If a normal green plant was crossed with the pale green plant, what would be the genotype of the first filial generation (F generation)? Show your working. (4 mks)
- (c) If heterozygous plants were self-pollinated and the resulting seeds planted, work out the proportion of their offspring that would grow to maturity. (2 mks)

2. The diagram below illustrates the structure of the kidney nephron.



- (a) Name the part labelled E.
- (b) How is the part labelled F adapted to its function?
- (c) State three physiological mechanisms of controlling the human body temperature during a cold day.
- (a) In an investigation, equal amounts of water was placed in three test tubes labelled G, H

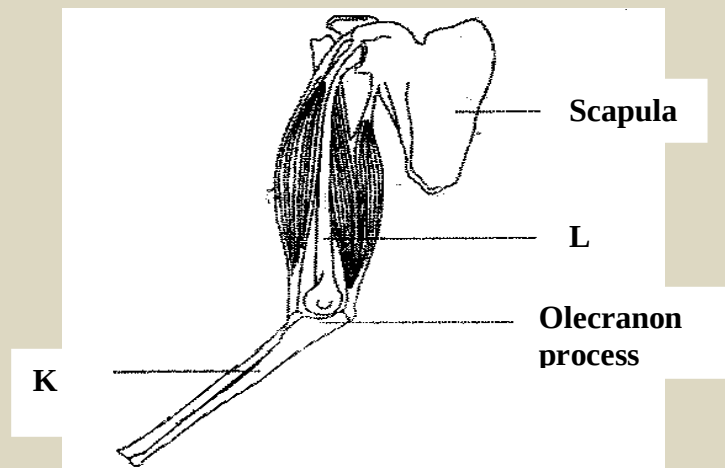
KCSE BIOLOGY PAPER 2 2006-2019 (MR CHEPKWONY)

and J. Pondweeds of equal length were dropped in each test tube. The test tubes were then placed in identical conditions of light and carbon(IV) oxide at different temperatures for five minutes. After five minutes, the bubbles produced in each test tube were counted for one minute. The results were as shown in the table below.

Test tube	Temperature (°C)	Number of bubbles
G	20	28
H	35	42
J	55	10

- (i) Name one requirement for this process that is not mentioned in the investigation. (1 mk)
- (ii) Name the gas produced in this investigation. (1 mk)
- (iii) Account for the results in test tubes H and J. (2 mks)
- (b) State two ways in which the human intestinal villus is adapted to its function. (4mks)

4. (a) The diagram below illustrates the arrangement of bones and muscles in the human arm.



- (i) Name the bones labelled K and L. (2mks)

K.....
L.....

- (ii) Explain how the upward movement of the lower arm is brought about by the bones and muscles shown diagram above. (3 mks)

- b) State three ways in which support is brought about in a leaf. (3 mk)

5. a) Describe the process of inhalation. (4mks)

b) Explain the mechanism of stomatal opening. (4mks)

SECTION B (40 mks)

Answer question 6 (compulsory) and either question 7 or 8 in the spaces provided

6 The data provided below represent populations of a predator and its prey over a fifty years period.

TIME IN YEARS	POPULATION IN RELATIVE NUMBERS	
	POPULATION OF P	POPULATION OF Q
5	24500	17000
10	30000	20500
15	33500	26000
20	33500	30000
25	31000	33000
30	27000	32000
35	25000	30000
40	29000	27500
45	32500	28000
50	34000	28500

(a) (i) Using the same axes, draw graphs of the relative populations of P and Q against time. (7 mks)

(ii) With a reason, identify the curve that represents the prey. (2 mks)

(iii) Account for the two populations between 25 and 32 years. (2 mks)

(iv) Which years were the two populations equal? (2 mks)

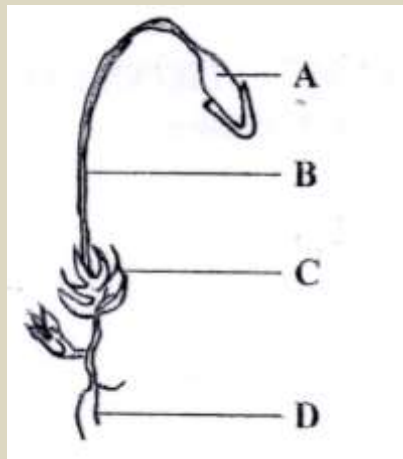
- (v) Apart from predation, state three biotic factors that may have led to the decline of the prey population. (3 mks)
- (b) Describe the hazards of air pollution by Sulphur (IV) Oxide. (4 mks)
7. Using a relevant example in each case, describe simple and conditional reflex actions. (20 mks)
- 8.
- (a) Using a relevant example, describe how an allergic reaction occurs in a human being. (10 mks)
- (b) Describe how environmental factors increase the rate of transpiration in terrestrial plants. (10 mks)

BIOLOGY
K.C.S.E PAPER 231/2 2013
QUESTIONS

SECTION A(40 mks)

Answer all the question in this section in the spaces provided

1. (a) The diagram below represents a plant in the division Bryophyta.



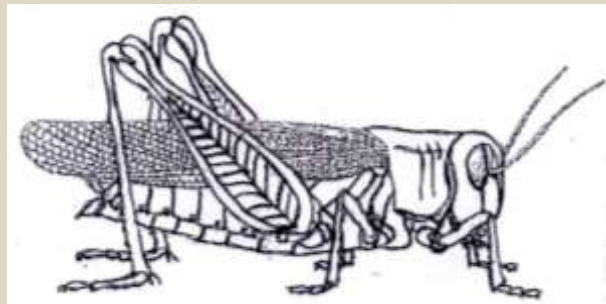
Name the parts labelled B and D.

(2 mks)

State one function for each of the parts labelled A and C.

- b) The diagram below represents a member of the kingdom Animalia.

(2 mks)



- i) Name the phylum to which the organism belongs.

(1 mk)

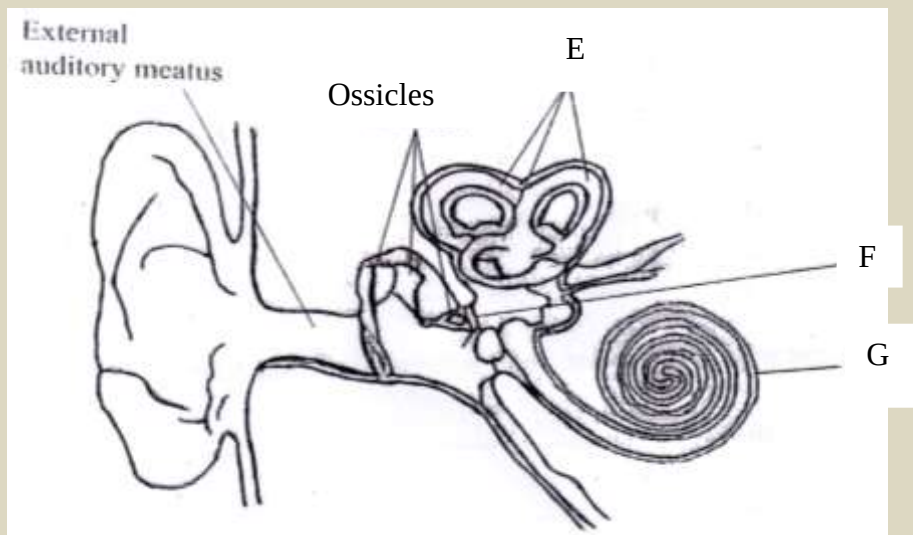
- ii) Using observable features in the diagram, give three reasons for the answer in b (i)

(3 mks)

2. The diagram below represents the human ear.

FOR MARKING External auditory
meatus

OR CALL 0724351706 Mr Chepkwony



- a) Name the labelled E, F, and G. (3 mks)
- b) How is each of the following adapted to its function?
- i) External auditory meatus; (2 mks)
 - ii) Ear ossicles (2 mks)
- c) Name one defect of the human ear. (1 mk)
3. a) Explain the importance of the following in photosynthesis; (3 mks)
- Light;
 - Carbon IV oxide;
 - Chlorophyll
- b) Name one appropriate food substance for each of the following enzymes:
- Ptyalin.
 - Pepsin..
- c) State the cause and two symptoms of Beri-beri.
- Cause..... (1 mk)
 - Symptoms..... (2 mks)

4.

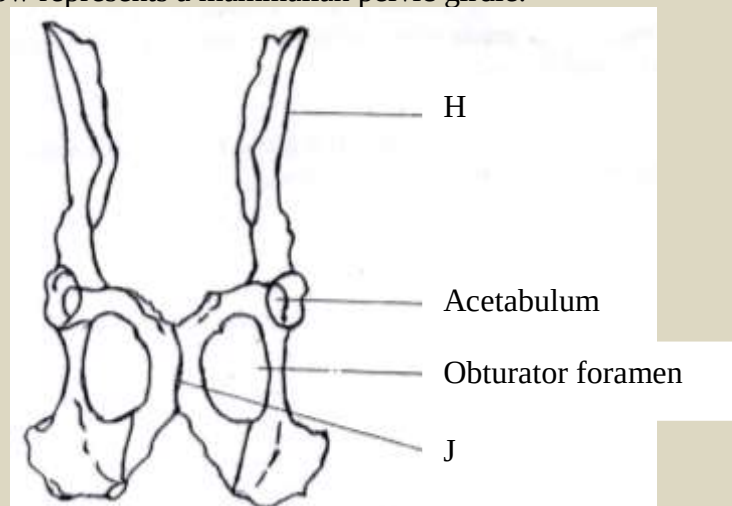
In an investigation, a variety of pea plants grown from seeds with smooth coats were crossed with plants grown from seeds with wrinkled coats. All the seeds obtained in the first filial (F) generation had smooth seed coats.

Using the letter R to represent the gene for smooth seed coat, work out the genotype of the F₁ generation. Show your working. (3 mks)

If the F₁ generation was selfed, determine the phenotypic ratio of the second filial (F₂) generation. Show your working. (3 mks)

If the total number of seeds in the F₂ generation was 14640, calculate the number of seeds with wrinkled coats. Show your working. (2 mks)

5. The diagram below represents a mammalian pelvic girdle.



a) How are the structures labelled H and J adapted to their function?

H..... (2 mks)

J..... (2 mks)

b) State the function of obturator foramen. (1 mk)

i) Name the bone that articulates with the pelvic girdle at acetabulum

ii) Name the type of joint formed by the acetabulum and the bone named in(c)(i) above.

c) Name the bone formed by the fusion of caudal vertebrae in human beings.

SECTION B (40 mks)**Answer question 6(compulsory) and either question 7 or 8 .**

6. A scientist carried out an investigation to find out the population growth of mice under laboratory conditions. Twenty young mice were placed in a cage. The results obtained from the investigation were as shown in the table below.

Time in months	0	2	4	6	8	10	12	16	18
Number of mice	20	20	65	115	310	455	450	145	160

On the grid provided, draw a graph of the number of mice against time.
account for the changes in mice population between

- 0 to 2 months (2 mks)
2 to 6 months (2 mks)
6 to 10 months (2 mks)
10 to 12 months (2 mks)

- i) Between which two months was the population change greatest? (1 mk)
ii) Calculate the rate of population change over the period in (c) (i) above. (2 mks)

What change in population would be expected if the investigation was continued to the 19th month? (1 mk)

To obtain the observed results state two variables that were kept constant during the investigation. (2 mks)

7. a) describe the process of blood clotting in human beings. (10 mks)
b) How are respiratory surfaces in mammals adapted to their functions? (10 mks)

8. Describe the role of the following organs in excretion and homeostasis.
the liver (10 mks)
b) The skin during hot environmental conditions (10 mks)

BIOLOGY
K.C.S.E PAPER 231/2 2014
QUESTIONS

SECTION A (40 mks)

Answer all the questions in this section in the spaces provided

1.

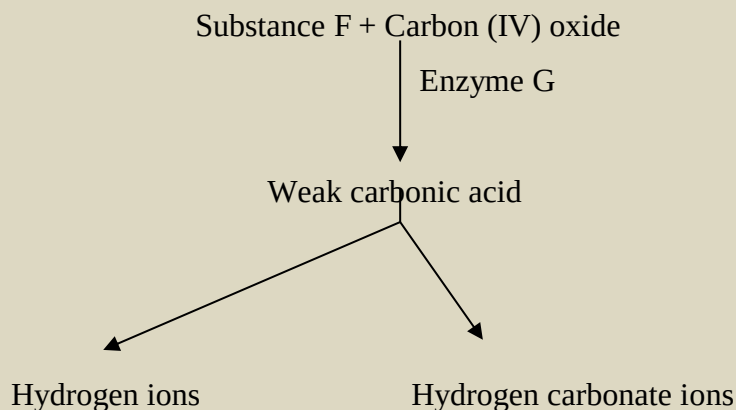
(a) State four characteristics of fruits dispersed by animals. (4 mks)

(b) State two roles of each of the following hormones in menstruation:

(i) Luteinising hormone (2 mks)

(ii) Oestrogen. (2 mks)

2. The diagram below illustrates the role played by red blood cells in the transportation of carbon (IV) oxide.



Other than carbon (IV) oxide transportation in the red blood cells, name the other form of carbon (IV) oxide transportation in humans. (1 mk)

(i) Name substance F (1 mk)

(ii) Name the enzyme mked G and state its role in the reaction.

Enzyme:

Role:

(2 mks)

(c) Explain the transportation of carbon (IV) oxide in red blood cells is advantageous (2 mks)

(d) Explain the role of calcium ions in blood clotting (2 mks)

KCSE BIOLOGY PAPER 2 2006-2019 (MR CHEPKWONY)

3. (a) Describe the mechanism of gaseous exchange in plants through the lenticels (3 mks)

(b) Explain each of the following:

(i) The tracheoles lack spiral bands of chitin; (3 mks)

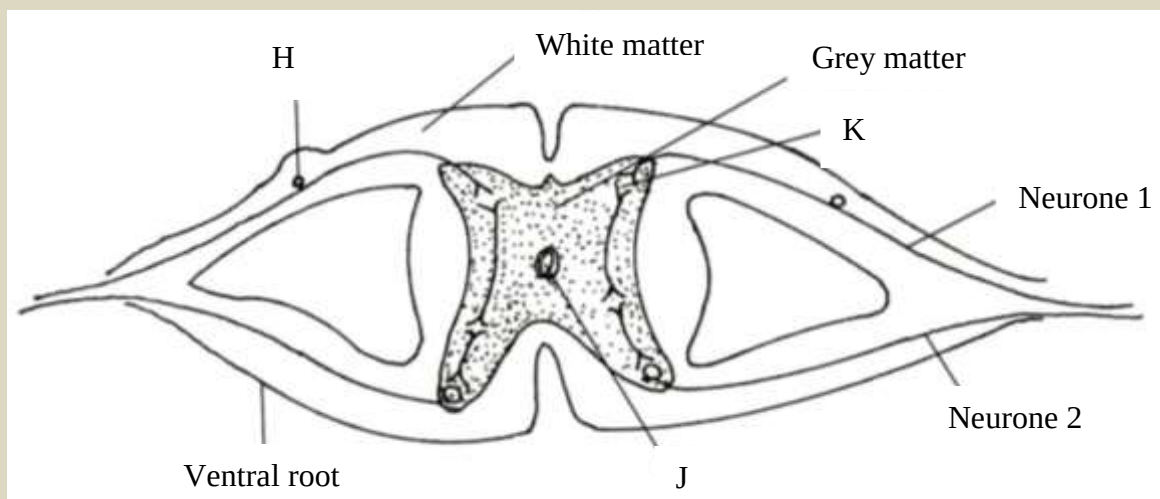
(ii) The floor of the mouth is lowered during inhalation in a bony fish (2 mks)

4. (a) How is sex determined in man? (4 mks)

(b) (i) Differentiate between sickle cell anaemia and sickle cell trait (2 mks)

(ii) Explain why people with sickle cell trait have an adaptive survival advantage over normal individuals in malaria endemic regions. (2 mks)

5. The diagram below represents the transverse section of the spinal cord.



(a) Name the part labelled H. (1 mk)

(b) State two functions of the fluid found in the part labelled J. (2 mks)

(c) Give a reason for the colour of the white matter. (1 mk)

(d) Name and give the function of the enzyme found at the part labeled K.

Name: (1 mk)

Function: (2 mks)

(e) On the diagram, use an arrow to show the direction of impulse transmission along the neurone labeled I. (1 mk)

SECTION B (40mks)

Answer questions 6(compulsory) and either question 7 or 8 in the spaces provided

6. An experiment was done to determine the uptake of nitrogen from the soil by broad bean seedlings. The experiment was done with one set of seedlings M grown in the atmosphere enriched with carbon (IV) oxide and another, set up of seedlings N grown in the normal atmosphere.

The amount of nitrogen in each seedling was measured in milligrams at intervals of ten days. The table below shows the results obtained.

	Amount of nitrogen in milligrams									
SET M	0	25	70	125	160	395	635	860	895	915
SET N	0	15	35	50	65	105	120	125	135	140
TIME (DAYS)	15	25	35	45	55	65	75	85	95	105

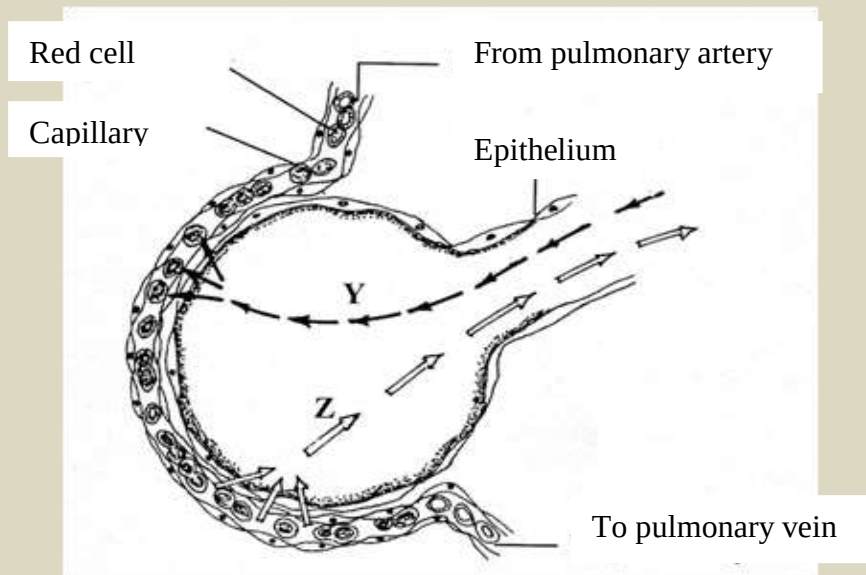
- (a) Using the same axis draw line graphs of nitrogen uptake by the two (M and N) sets of broad bean seedlings against time. (8 mks)
- (b) Determine the rate of uptake of nitrogen in Set M between 65 and 85 days. (2 mks)
- (c) i) What is the relationship between carbon (IV) oxide concentration in the air and nitrogen uptake?
ii) Account for the relationship in (c) (i) above (3 mks)
- (d) i) What would happen to the concentration of nitrogen in the seedlings in set M, if after 75 days the seedlings are transferred to a normal atmosphere (1mk)
ii) Explain your answer in (d) (i) above (2 mks)
- (e) State three ways in which nitrogen fixation occurs (3 mks)
- 7a) Explain how each of the following factors affects the rate of photosynthesis:
(i) Temperature; (2 mks)
(ii) Chlorophyll concentration (2 mks)
- b) Describe the process of carbohydrate digestion in human beings (16 mks)
8. a) How does excretion take place in plants? (4 mks)
b) Describe the role of the human skin in homeostasis (16 mks)

BIOLOGY
K.C.S.E PAPER 231/2 2015
QUESTIONS

SECTION A (40 mks)

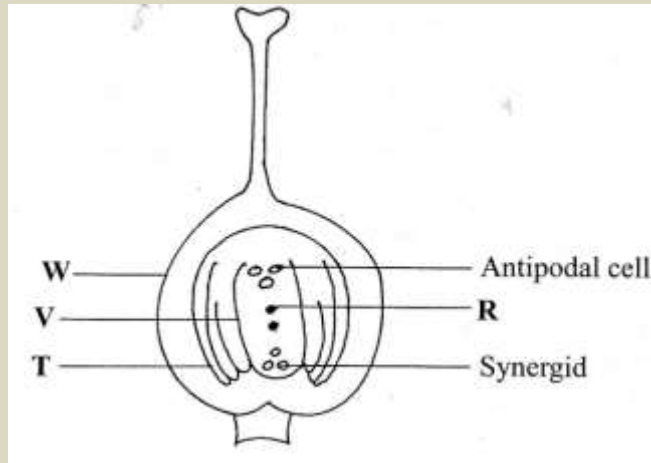
Answer all the questions in this section in the spaces provided

1. The diagram below illustrates a blood capillary surrounding a structure for gaseous exchange in human beings.

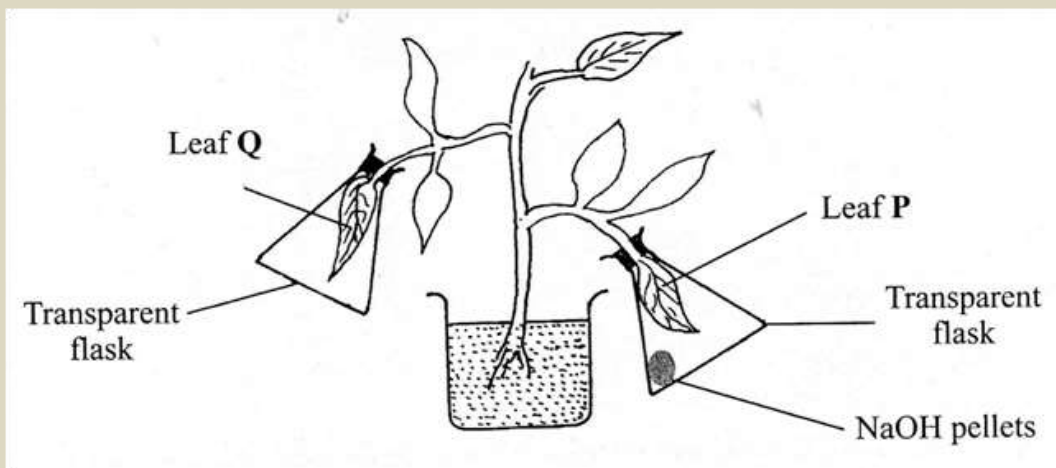


- a) Name the gaseous exchange structure (1mk)
- b) Identify the gases labeled Y and Z (1 mk)
- Y (1 mk)
- Z (1 mk)
- c) How does the gas labeled Y reach the inside of the blood capillary? (3mks)
- d) How does cigarette smoking lead to lung cancer? (2 mks)

2. The diagram below illustrates the structure of the female part of a flower

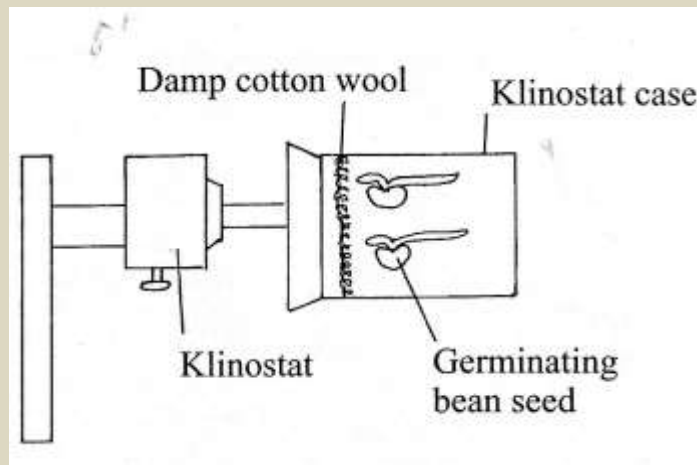


- a) Name the part labeled W (1 mk)
- b) Describe what happens when the pollen tube enters the structure labeled V (5 mks)
- c) What do the structures labeled R and T develop into after fertilization?
R (1 mk)
T (1 mk)
3. a) What is meant by the term genetics? (1 mk)
- b) State two examples of discontinuous variation (2 mks)
- c) A female which sickle cell trait marries a normal man. The allele for sickle cell is Hbs and the normal allele is HbA. Determine the probability that their first born will have the sickle cell trait. Show your working. (5mks)
4. In an experiment to investigate a factor affecting photosynthesis, a potted plant which had been kept in the dark overnight was treated as shown in the diagram below and exposed to light.



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- a) Why was the potted plant kept in the dark overnight? (1 mk)
- b) Which factor was being investigated in the experiment? (1 mk)
- (i) Which test did the students perform to confirm photosynthesis in the leaves labeled P and Q? (1 mk)
- (ii) State the results obtained in the leaves labeled P and Q (1 mk)
- (iii) Explain the results obtained in the leaves labeled P and Q (1 mk)
- (iv) What was the purpose of leaf Q in the experiment? (1 mk)
5. In an experiment to investigate a plant response, the set up shown in the diagram below was used.

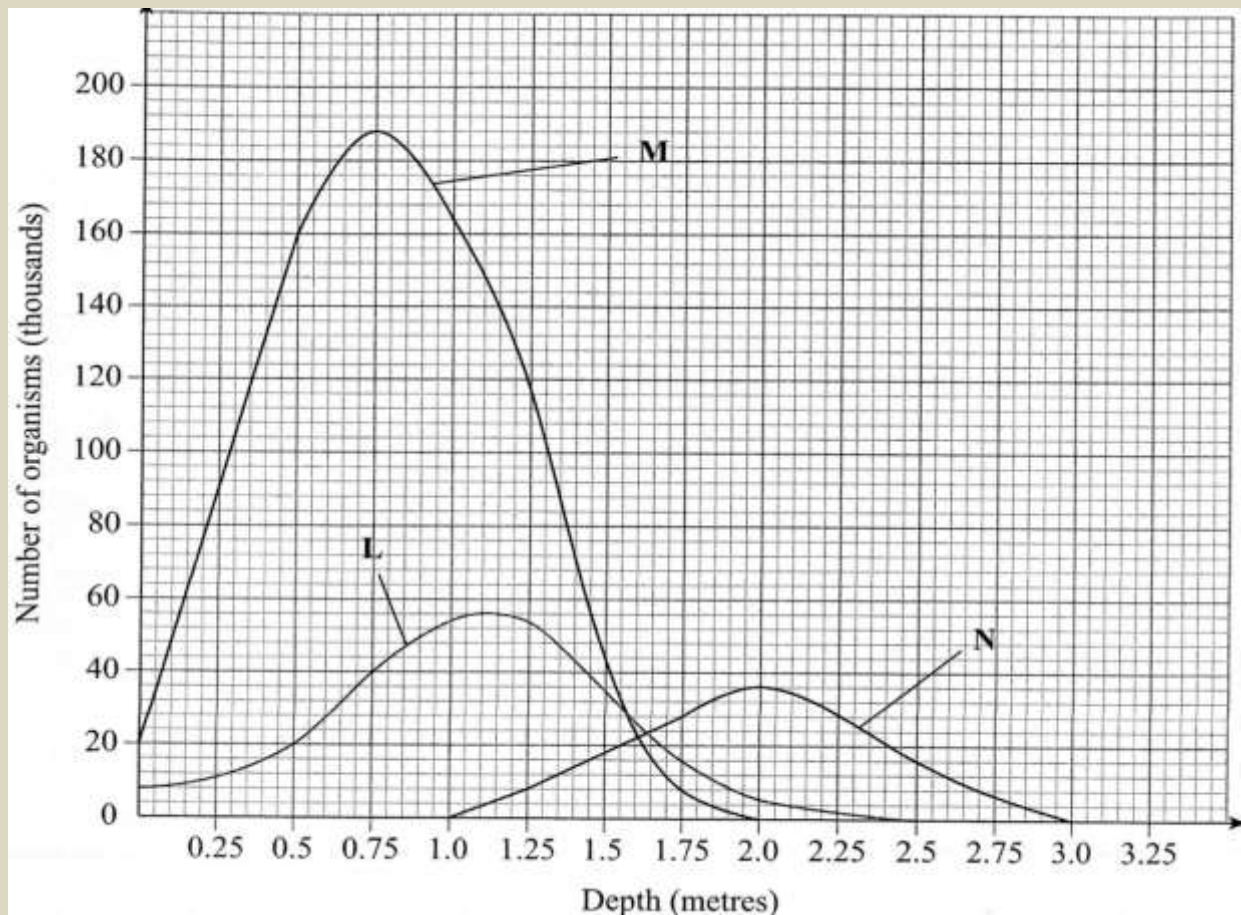


- a) Name the type of response that was being investigated (1 mk)
- b) If the klinostat was not rotating:
- i) State the observations that would be made on the seedlings after three days (2 mks)
- ii) Explain the observations in (b) (i) above (3 mks)
- c) If the experiment was repeated with the klinostat rotating
- i. State the observation that was made on the seedlings after three days; (1 mk)
- ii. Give a reason for the observation made on the seedlings (1 mk)

SECTION B (40 MKS)

Answer question 6 (compulsory) and either question 7 or 8 in the spaces provided

6. The graph below shows the relative numbers of three main species of organisms in a pond



- a) Giving a reason for your answer, which of the species is a
- (i) Producer? (1 mk)
Reason (1mk)
- (ii) Secondary consumer? (1mk)
Reason (1 mk)
- b) State the depths at which each of the populations labeled L, M and N is at its optimum.
- L (1 mk)
M (1 mk)
N (1mk)
- c) (i) Which method may have been used to determine the population of organisms labeled N in the pond? (1 mk)
- (ii) Give a reason for your answer in (c) (i) above (1mk)

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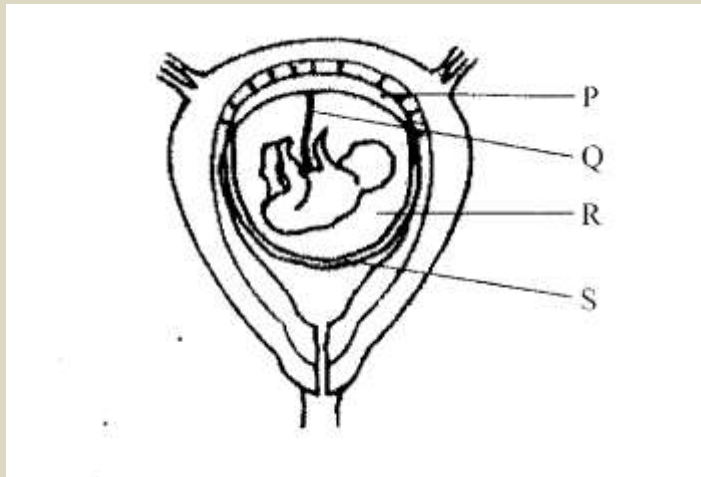
- (iii) State the assumptions made when using the method in (c) (i) above (4 mks)
- d) State two reasons why primary productivity in the pond decreases with depth. (2 mks)
- e) Explain the ecological importance of fungi to plants (2 mks)
- f) Why is flooding likely to lead to a cholera outbreak? (3mks)
7. Explain the various ways in which seeds and fruits are adapted to dispersal (20 mks)
8. How is a mammalian heart structurally adapted to its function? (20 mks)

BIOLOGY
K.C.S.E PAPER 231/2 2016
MARKING SCHEME

SECTION A (40 MARKS)

Answer all the questions in this section in the spaces provided

1. The diagram below represents a human foetus in a uterus

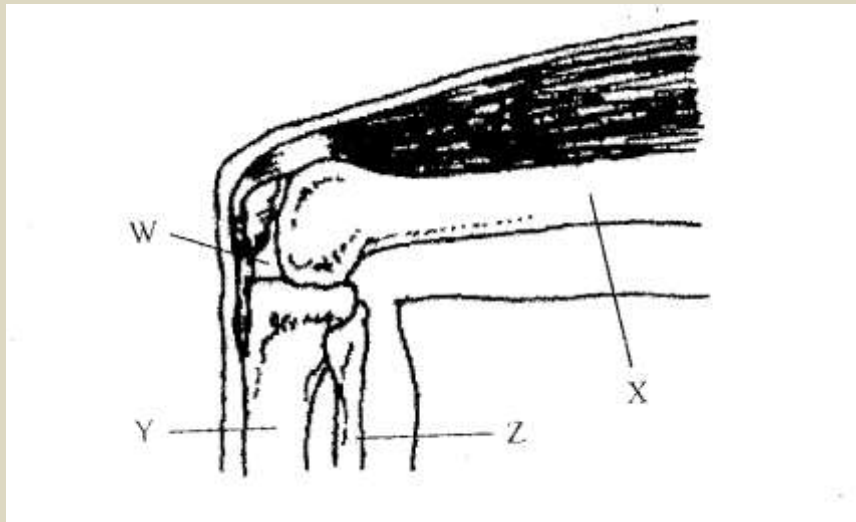


- a) Name the part labeled S. (1mks)
- b) i) Name the types of blood vessels found in the structure labeled Q. (2 mks)
ii) State the difference in composition of blood found in the vessels named in (b) (i) above (2mks)
- c) Name two features that enable the structure labeled P to carry out its function. (2 mks)
- d) State the role of the part labeled R. (1mk)

2. a) i) How is sex determined in man? (2 mk)
- ii) Explain why people with sickle cell trait have an adaption survival advantage over normal individuals in malaria endemic regions. (2 mks)

KCSE BIOLOGY PAPER 2 2006-2019 (MR CHEPKWONY)

3. The diagram below represents bones at a joint found in the hind limb of a mammal.



a) Name the bones labeled X, Y and Z (3 mks)

X

Y

Z

Bi) Name the substance found in the place labeled w. (1 mk)

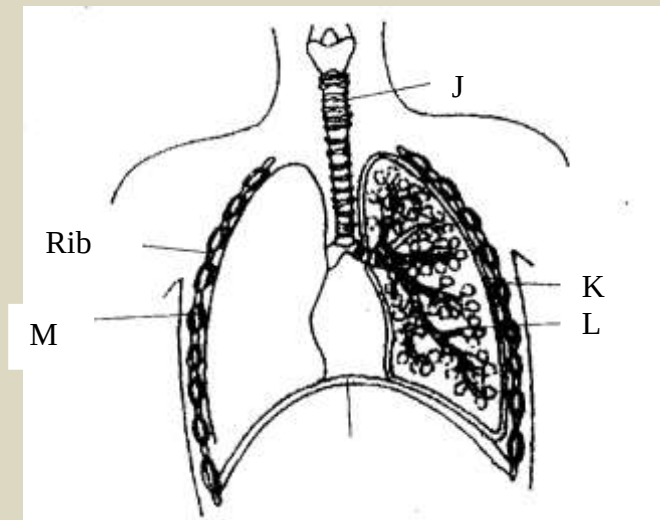
ii) State the function of the substance named in (b) (i) above (1 mks)

c) Name the structure that joins bones together at the joint. (1 mk)

d) State the difference between ball and socket joint and the one illustrated in the diagram above. (1mks)

e) Name the structure at the elbow that performs the same function as the patella. (1mks)

4. The diagram below represents some gaseous exchange structures in humans.



- a) Name the structures labeled K, L and M. (3 mks)
- b) How is the structure labeled J suited to its function ? (3 mks)
- c) Name the process by which inhaled air moves from the structure labeled L into blood capillaries (1 mks)
- d) Give the scientific name of the organism that causes tuberculosis in humans (1 mks)

5. A freshly obtained dandelion stem measuring 5cm long was split lengthwise to obtain two similar pieces. The pieces were placed in two different solutions of different concentration in petri-dishes (L_1 and L_2) for 20 minutes. The appearance after 20 minutes is as shown.



- a) Account for the appearance of the piece in solutions L_1 and L_2 . (6 mks)
- b) State the significance of the biological process involved in the experiment (2 mks)

SECTION B (40 mks)

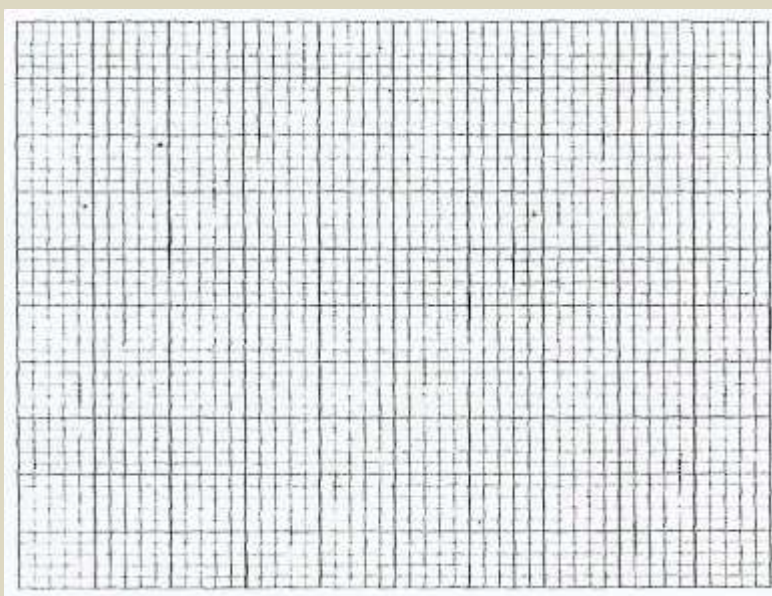
KCSE BIOLOGY PAPER 2 2006-2019 (MR CHEPKWONY)

Answer question 6 (compulsory) and either question 7 or 8

6. An experiment was carried out to investigate the effect of temperature on the rate of reaction catalysed by an enzyme. The results are shown in the table below

Temperature (°C)	Rate of reaction in mg of products per unit time
5	0.2
10	0.5
15	0.8
20	1.1
25	1.5
30	2.1
35	3.0
40	3.7
45	3.4
50	2.8
55	2.1
60	1.1

- a) On the grid provided plot the rate of reaction against temperature.



- b) When was the rate of reaction 2.6 mg of product per unit time? (2 mks)

- c) Account for the shape of the graph between

- i) 5°C and 40°C

(2 mks)

KCSE BIOLOGY PAPER 2 2006-2019 (MR CHEPKWONY)

ii) 45°C and 60°C

(3 mk)

d) Other than temperature name two ways in which the rate of reaction between 5°C and 40°C could be increased

(2 mks)

e i) Name one digestive enzyme in the human body which works best in acidic condition.

(1 mks)

ii) How is the acidic condition for the enzyme named in (e) (i) above attained?

(2 mks)

f) The acidic condition in (e) (iii) above is later neutralized

i) Where does the neutralization take place?

(1mk)

ii) Name the substance responsible for the neutralization

(1 mk)

7. Using a relevant example in each case, describe simple and conditional reflex actions

(20 mks)

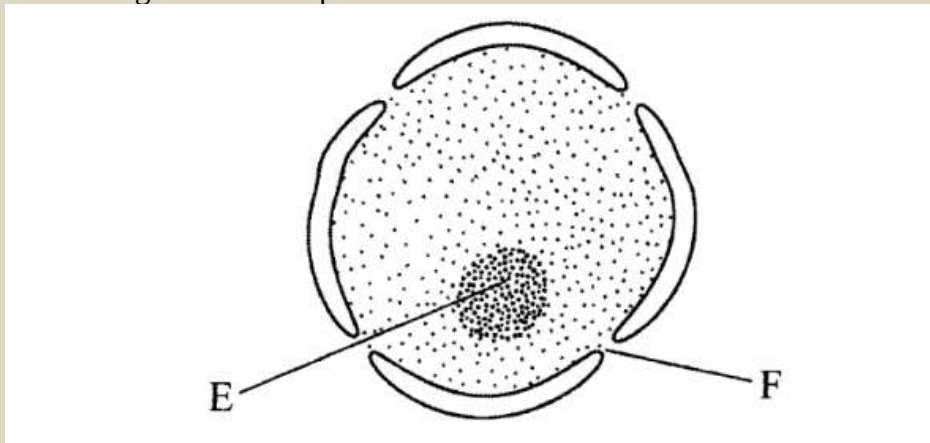
8. Describe how the mammalian heart is structurally adapted to its function

(20 mks)

KCSE 2017 BIOLOGY PAPER 2

Answer all the questions in this section in the spaces provided.

1. The diagram below represents a nucleus.



(a) Name the structures labelled E and F. (2 marks)

(i) E

(ii) F

(ii) State the function of F. (1 mark)

With reference to the nucleus state one difference between an animal and a bacterial cell.

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(1 mark) (b) Name the plant cell organelle:

(i) that stores chlorophyll (1 mark)

(ii) responsible for intracellular digestion. (1 mark)

(c) State two main functions of the vacuole in the amoeba. (2 marks)

2. The table below shows variations in the form carbon (IV) oxide is transported in the blood at rest and during physical exercise.

Carbon (IV) oxide transport in blood plasma at rest and during exercise		
Form of transport	Rest (Mol/l)	Exercise (Mol/l)
Dissolved carbon (IV) oxide	0.52	0.97
Bicarbonate ion	12.34	13.68
Carbon (IV) oxide bound to protein	0.26	0.16
Total carbon (IV) oxide in plasma	13.12	14.81
pH of blood	7.42	7.09

(a) Explain why more carbon (IV) oxide is transported in the form of bicarbonate ion. (2 marks)

(b) Account for the high total plasma content of carbon (IV) oxide during exercises. (3 marks)

(c) State how one's involvement in the exercises affects blood pH. (2 marks)

(d) Name the protein responsible for the transport of carbon (IV) oxide in the blood. (1 mark)

3. The diagram below illustrates the appearance of a plant cell after it had been put in a certain solution.



- (a) Explain the appearance of the cell at the end of the treatment. (3 marks)
(b) Explain the results obtained if a red blood cell is subjected to the same treatment. (3 marks)

(c) Explain why transfusion with distilled water is not recommended for a dehydrated patient. (2 marks)

4. (a) Explain how the sex of a male child is determined in human beings. (2 marks)

(b) (i) Define the term diploidy. (1 mark)

(ii) Name the type of cell division that gives rise to diploid cells. (1 mark)

(iii) Name the type of cells in which the process named in (b) (ii) above occurs. (1 mark)

(iv) State the significance of diploidy. (2 marks)

(c) Name the hormone responsible for the development of secondary sexual characteristics in human males. (1 mark)

5. In beans, the gene for purple colour is dominant over the gene for White colour. A pure breeding bean plant with purple colour was crossed with a heterozygous bean plant.

(a) Using the letter P to represent the gene for purple colour, Work out the genotypic ratio of the offspring. (5 marks)

(b) State two advantages of using genetically modified varieties in bean farming. (2 marks)

(c) State how in-breeding leads to reduced hybrid vigour. (1 mark)

Section B (40 marks) Answer question 6 (compulsory) and either question 7 or 8 in the spaces provided after question 8.

6. In an investigation, two potted plants G and H belonging to the same species were exposed to increasing light intensities at different temperatures, 30 °C and 20°C respectively. The rate of 'h 'n the table below:

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Light intensity (in arbitrary units)	1	2	3	4	5	6	7	8
Rate of photosynthesis for plant G at 30°C	0	84	148	196	232	260	284	296
Rate of photosynthesis for plant H at 20°C	0	72	115	148	170	186	204	216

- (a) On the same axis, plot graphs of rate of photosynthesis against light intensity for plants G and H (8 marks)
- (b) State the aim of the investigation. (1 mark)
- (c) Account for the difference in the rate of photosynthesis in the two plants. (3 marks)
- (d) Account for the difference in the rate of photosynthesis in the two plants between the following light intensities:
- (i) 1—4 units (2 mark)
- (ii) 4—8 units.(2 mark)
- (e) (i) Predict the rate of photosynthesis at light intensity of 16 units.(1 mark)
- (ii) Give a reason for your answer in (e) (i) above.(1 mark)
- (f) State one internal and one external factor that could be limiting in the investigation. (2 marks)
7. Explain the importance of protecting the forest ecosystem with reference to the following:(20 marks)
- (a)climate change
- (b)biodiversity
- (c)biotechnology
- (d)water conservation
- (e)pollution.
8. Describe how the mammalian eye is structurally adapted to its function. (20 marks)

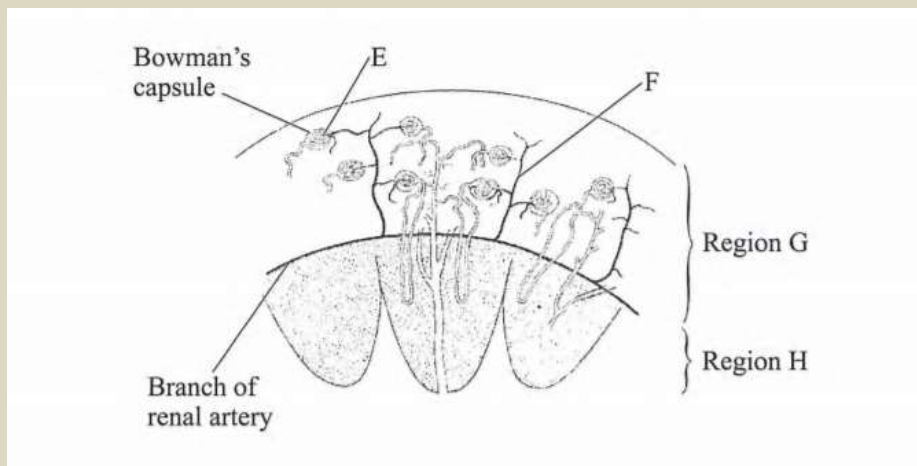
KCSE 2018 BIOLOGY PAPER 2

Answer all the questions in the spaces provided.

SECTION A (40 marks)

Answer all the questions in this section in the spaces provided.

1. The diagram below illustrates a section of the mammalian kidney.



(a) Name the structures labelled E and F. Region H (1 marks)

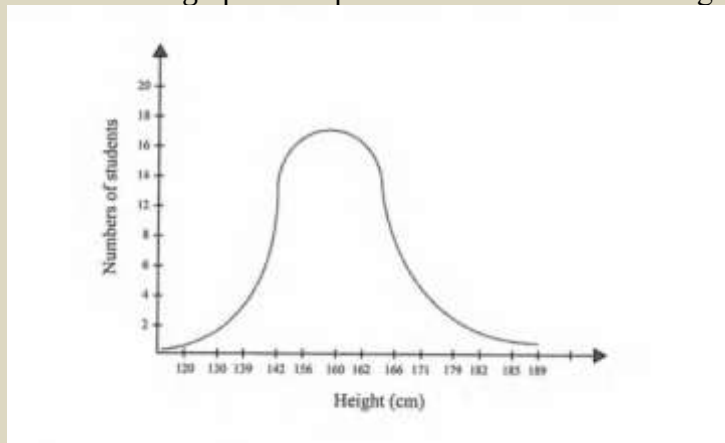
E(1 marks)

F.....(1 marks)

(b) Explain the processes that take place in the regions labelled G and H. Region G(3 marks)

Region H(3 marks)

2. Below is a graphical representation of students' height in a classroom.



(a) Name the type of curve illustrated.

(b) (i) State the type of variation represented by the curve.

(ii) State two meiotic processes that lead to variation among organisms. (2 marks)

(iii) Explain the role of variation in organisms.(2 marks)

(c) EKplain the need for genetic counselling in present day health facilities. (2 marks)

3.The amount of blood flowing through certain parts in the mammalian body at different activity levels was measured and results tabulated as shown in the table below.

Parts of the body	Blood flow (cm ³ /minute)		
	At rest	During light exercises	During strenuous exercise
Alimentary canal	1,100	780	350
Cardiac muscles	100	200	1,300
Skeletal muscles	900	4600	15,000

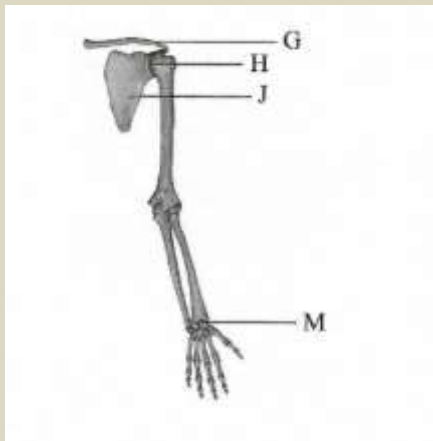
(a) Account for:

(i) the high blood-flow through the cardiac and skeletal muscles during strenuous exercises. (4 marks)

(ii) the results obtained for the alimentary canal at rest. (2 marks)

(b) Name two waste materials excreted by both the skin and the kidneys. (2 marks)

4.The diagram below illustrates the arrangement of bones in a human arm.



(a) Name:

(i) G

the type of joints formed at points H and M.

H (1 mark)

M (1 mark)

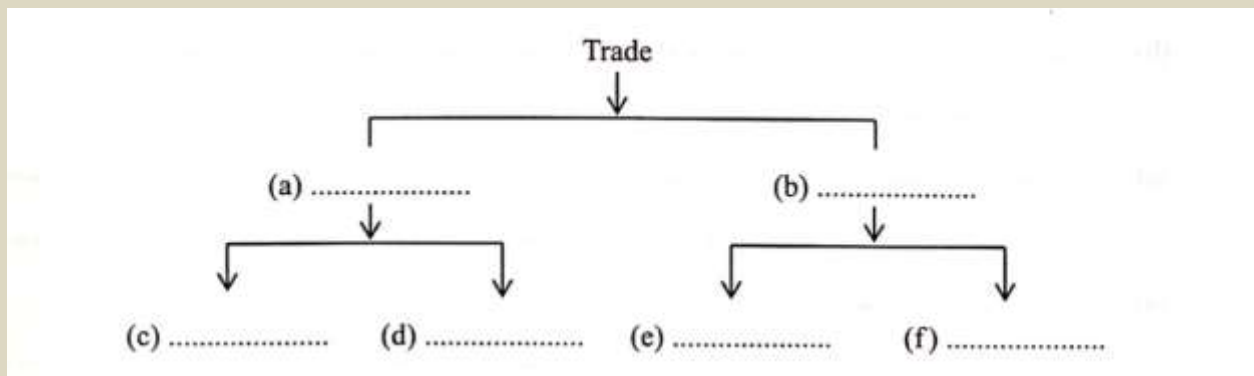
(ii) bone G (1 mark)

(b) Name bone J and state how it is adapted to its functions. Name: (1 mark)

Adaptation (3 marks)

(c) State one functional difference between a tendon and a ligament. (1 mark)

5. The photograph below represents a plant in a certain Division.



- (a) (i) Name the Division to which the plant belongs.(1 marks)
 (ii) With reference to the photograph, state three observable features of the Division named in a(i) above. (3 marks)

(b) Name the parts labelled N and P.

N.....(1 marks)

P.....(1 marks)

(c) Explain how the part labelled Q is adapted to its functions.(2 marks)

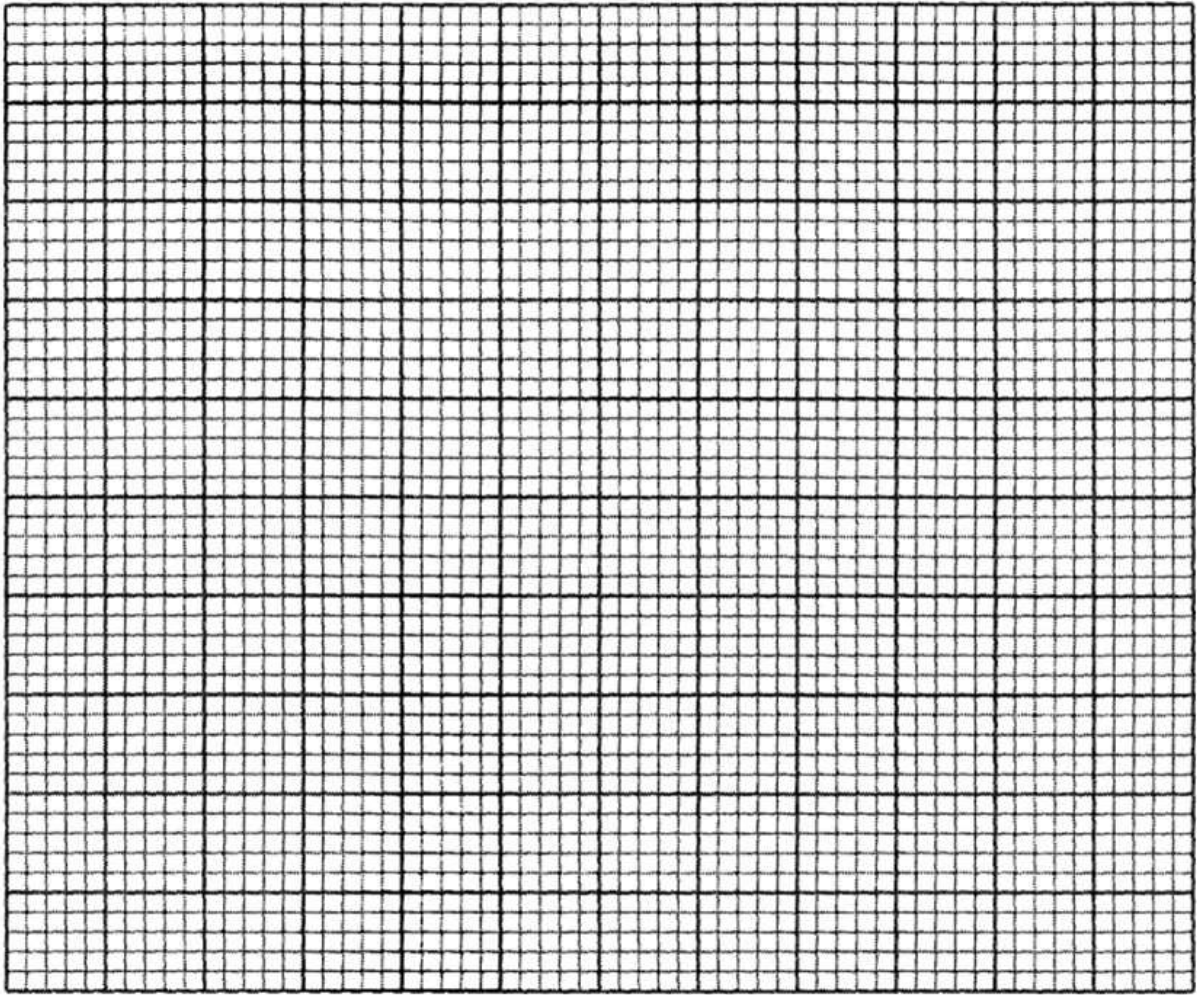
SECTION B (40 marks)

Answer question 6 (compulsory) and either question 7 or 8 in the spaces provided after question 8.

6. The effect of auxin concentration on growth response of two parts of a plant, X and Y was investigated over a period of time. The results were tabulated as shown in the table below.

Concentration of Auxin (in parts per million)	10^{-6}	10^{-5}	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1	10^1	10^2
Percentage inhibition /stimulation on part X	0	40	55	40	0	-45	-90	0	0
Percentage inhibition /stimulation on part Y	0	0	0	25	65	155	210	125	-25

- (a) On the same axis, draw line graphs of the effect on growth of the two parts, X and Y (percentage inhibition or stimulation) against the concentration. (8 marks)



(b) With reasons, name the two parts of the plant, X and Y. X.....(1 marks)

Reason.....(1 marks)

Y.....(1 marks)

Reason.....(1 marks)

(c) From the graph identify: (i) the point at which the percentage stimulation was the same for both X and Y. (1 marks)

(ii) the optimum concentration of auxins required for part Y (1 marks)

(d) State three ways in which the effects of auxins on plants is applied in flower farming.(3 marks)

(e) Distinguish between simple and conditioned reflex action (3 marks)

7. (a) Describe the mode of reproduction in a named fungus.(5 marks)

(b) Describe the role of hormones in the human menstrual cycle. (15 marks)

8. Describe what happens to a meal rich in proteins along the alimentary canal from ingestion to egestion. (20 marks)

KCSE 2019 BIOLOGY PAPER 2

Answer all the questions in the spaces provided.

1. In an experiment to investigate the effect of Sodium chloride on the growth rate in a spinach seedling, seeds were treated with different concentrations of sodium chloride. The results are as recorded in the table below.

Concentration of sodium chloride (mol/l)	Percentage of spinach seeds which started to grow roots	Mean root length (mm)
0.00	99.98	17.70
0.06	98.20	15.60
0.12	92.0	10.20
0.18	54.0	7.60

(a) From the results in the table above, explain the effect of increasing the concentration of sodium chloride. (3 marks)

(b) Apart from a ruler, state two other equipment one would need to determine the rate of growth in the roots. (2 marks)

(c) With a reason, state one other part of the seedling the students would focus on to determine the effect of sodium chloride on growth. (2 marks)

(d) State the likely effect on the seedling of increasing the seedling of increasing the concentration of sodium chloride to 2.20 mol/l. (1 mark)

2. The table below shows results of blood cell counts per mm^3 of blood from a sample of people living at different altitudes.

Red blood cells ($\times 10^4$)	4.8	5.3	6.7	7.6	8.47	9.82
White blood cells ($\times 10^4$)	0.45	0.45	0.45	0.45	0.45	0.45
Altitude (metres)	750	1,500	2,250	3,000	4,500	4,500

(a) Explain the relationship between:

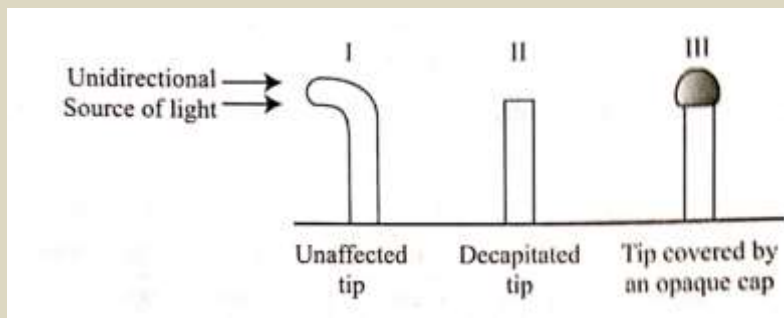
(i) Red blood cells count and the altitude;(3 mark)

(ii) White blood cells count and the altitude.(3 mark)

(b) Explain why chances of nose-bleeding increase with altitude in humans. (2 marks)

3. (a) State one importance of irritability to living organism. (1 mark)

(b) In an experiment, students treated seedlings as illustrated below.



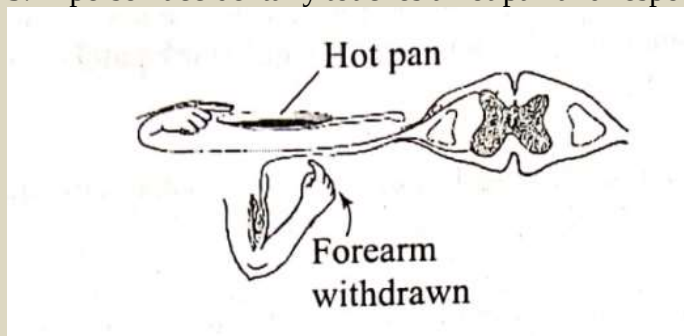
- (i) Account for the observations made in seedling I (3 marks)
(ii) Explain the similarity in the end results made in seedlings II and III (2 marks)
(iii) State the likely treatment that would make seedlings II and III respond like seedling I. (2 marks)

4. In cats, the gene for fur colour is sex-linked. Letter G represents the gene for ginger fur colour while letter B represents the gene for black fur colour in a given cat species. These genes are codominant.

Heterozygous females have ginger and black patches of fur and their phenotype is described as tortoise-shell.

- (a) With reference to the information given above, what is meant by the term codominance? (1 mark)
(b) Explain why male cats with a tortoise-shell phenotype do not usually occur. (2 marks)
(c) A tortoise-shell female was crossed with a black male. Determine the genotypes and phenotypes of the offspring. (5 marks)

5. A person accidentally touches a hot pan and responds as illustrated in the diagram below.



- (a) Explain how the response illustrated above occurs. (6 marks)
(b) Explain how auxins are utilised as selective weed killers in agriculture. (2 marks)

SECTION B (40 Marks)

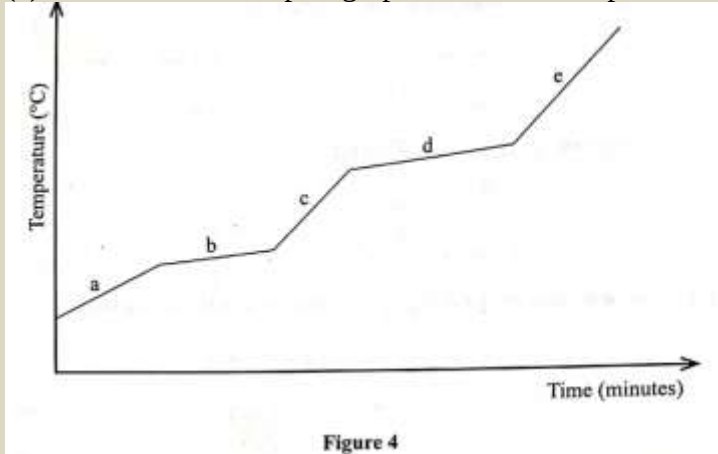
Answer question 6 (compulsory) and either 7 or 8

KCSE BIOLOGY PAPER 2 2006-2019 (MR CHEPKWONY)

6. The table below shows the rate of product formation for two enzymes, H and J over a range of pH values.

pH	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
Rate of product formation for enzyme H (mg/hr)	34.5	40.5	33.5	15.0	–	–	–	–	–	–
Rate of product formation for enzyme J (mg/hr)	–	–	–	15.0	20.0	30.0	40.5	23.5	11.0	6.0

(a) On the same axis, plot graphs of the rate of product formation against pH. (8 marks)



(b) Account for the rate of product formation for enzyme H between:

(i) pH 1.0 and 3.0 (3 marks)

(ii) pH 3.0 and 7.0. (3 marks)

(c) From the graph, determine:

(i) The pH value at which the rate of product formation of the two enzymes was the same (1 mark)

(ii) The value of the rate of product formation for enzymes H and I at the pH value stated in (G)(i) above (1 mark)

(iii) The optimum pH value for enzyme J (1 mark)

(d) State one variable that may lead to the change in the optimum rate of product-formation of the two enzymes. (1 mark)

(e) Suggest with a reason, the likely part of the human alimentary canal where enzyme H would be found. (2 marks)

7. Giving examples, describe the following interactions among organism: (20 marks)

8. Explain the effects of increased physical activity on the following organ systems: (20 marks)

(a) Heart

(b) Lungs

(c) Kidneys

(d) Skin.

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